Measure Theory Notes

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1 Motivation: from Riemann to Lebesgue

Let $f : [a, b] \to \mathbb{R}$ be bounded. The Riemann integral is constructed by dividing the domain $[a, b]$ into finitely many subintervals $[t_{k-1}, t_k]$ and approximating the area under the curve with sums

$$\sum_{k=1}^{n} f(\xi_k)(t_k - t_{k-1}), \qquad \xi_k \in [t_{k-1}, t_k].$$

When these sums converge as the partition is refined, f is said to be Riemann-integrable.

This construction has two limitations:

- Functions with many discontinuities are not Riemann-integrable, even though intuitively they should have an area. The classical example is $\mathbf{1}_{\mathbb{Q}\cap[0,1]}$.
- The theorems on interchanging limit and integral require strong hypotheses.

An earlier attempt (Jordan, 1892) was to measure a bounded set E by approximating it from outside with *finite* covers by intervals:

$$\bar{c}(E) = \inf \left\{ \sum_{k=1}^{n} \ell(I_k) : E \subseteq I_1 \cup \cdots \cup I_n \right\},$$

where ℓ denotes length. This *Jordan content* fails on dense sets: every finite cover of $\mathbb{Q} \cap [0, 1]$ by open intervals is an open set that contains a dense subset of $[0, 1]$, and its closure (which only adds finitely many endpoints) contains $[0, 1]$; the sum of the lengths is then ≥ 1, so that $\bar{c}(\mathbb{Q} \cap [0, 1]) = 1$ and, by the same argument, $\bar{c}(\mathbb{Q}^c \cap [0, 1]) = 1$: two disjoint sets whose union is $[0, 1]$ would have "measures" adding up to 2. The decisive correction was to admit *countable* covers: with them $\mathbb{Q} \cap [0, 1]$ is covered by intervals of arbitrarily small total length (an interval of length $\epsilon/2^n$ around the n-th rational), and its measure turns out to be 0.

Lebesgue's idea (1904). Instead of partitioning the domain, the range is partitioned. Take a partition $a_0 < a_1 < \cdots < a_n$ of the range of f, and for each k consider the set

$$A_k = \{t \in [a, b] : a_{k-1} < f(t) \leq a_k\}.$$

The integral is approximated by

$$\sum_{k=1}^{n} a_k \cdot m(A_k),$$

where $m(A_k)$ is a notion of *measure* of the set A_k.

The goal is to construct, starting from a reasonable class of subsets called *measurable*, a function $m : \mathcal{A} \to [0, \infty]$ with properties such as:

- $A \subseteq B \implies m(A) \leq m(B)$.
- $m(A \cup B) = m(A) + m(B)$ when $A \cap B = \emptyset$.
- On $\mathbb{R}$, $m([a, b]) = b - a$ and m is translation invariant.

It will be seen that *not* every subset of $\mathbb{R}$ admits such a measure; there are subsets that are not measurable.

2 σ-algebras

Definition 2.1 – σ-algebra

Let X be a set. A family $\mathcal{A} \subseteq \mathcal{P}(X)$ is a σ-algebra on X if:
(i) $\emptyset \in \mathcal{A}$.
(ii) Closed under complements: $A \in \mathcal{A} \implies A^c \in \mathcal{A}$.
(iii) Closed under countable unions: $(A_n) \subset \mathcal{A} \implies \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.
The elements of $\mathcal{A}$ are called *measurable sets* (with respect to $\mathcal{A}$).

Remark. By De Morgan, a σ-algebra is also closed under countable intersections: if $(A_n) \subset \mathcal{A}$, then

$$\bigcap_n A_n = \left(\bigcup_n A_n^c \right)^c \in \mathcal{A}.$$

Moreover, $X = \emptyset^c \in \mathcal{A}$.

2.1 Examples

Example 2.1 – Trivial σ-algebra

$\mathcal{A} = \{\emptyset, X\}$ is the smallest possible σ-algebra on X. It satisfies the three axioms by inspection.

Example 2.2 – Discrete σ-algebra

$\mathcal{A} = \mathcal{P}(X)$ is the largest σ-algebra, where every subset is measurable. On small sets this is the natural choice.

Example 2.3 – Cofinite sets: an algebra that is not a σ-algebra

Let X be infinite and $\mathcal{A} = \{A \subseteq X : A \text{ finite or } A^c \text{ finite}\}$. This family is an *algebra* (closed under finite unions and intersections and under complements) but it is *not* a σ-algebra.

Example 2.4 – σ-algebra of coin sequences

Let $X = \{0,1\}^{\mathbb{N}}$ be the space of infinite sequences of zeros and ones (the results of tossing a coin infinitely many times, with 1 = heads). For each n define the event

$$E_n = \{x \in X : x_n = 1\} \qquad (\text{“the } n\text{-th toss is heads”}),$$

and take $\mathcal{A} = \sigma(E_1, E_2, \dots)$. This σ-algebra contains all the events described by countably many tosses: for example $\bigcap_n E_n$ (“always heads”), $E_1 \cap E_2^c$ (“heads and then tails”), or the limits of the next subsection. This space will reappear when studying measures and the Borel-Cantelli lemmas.

Proposition 2.1 – Intersection of σ-algebras

If $(\mathcal{A}_i)_{i\in I}$ is an arbitrary family of σ-algebras on a set X, then the intersection

$$\mathcal{B} = \bigcap_{i\in I} \mathcal{A}_i$$

is again a σ-algebra on X.

Proof.
The three axioms that define a σ-algebra are verified.

It contains the empty set. For each $i \in I$, $\mathcal{A}_i$ is a σ-algebra and therefore $\emptyset \in \mathcal{A}_i$. Since this holds for every i, $\emptyset \in \mathcal{B}$.

Closed under complements. Let $A \in \mathcal{B}$. For each $i \in I$, $A \in \mathcal{A}_i$, and since $\mathcal{A}_i$ is closed under complements, $A^c \in \mathcal{A}_i$. Therefore A^c belongs to all the $\mathcal{A}_i$, that is, $A^c \in \mathcal{B}$.

Closed under countable unions. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{B}$ be a sequence. Fix $i \in I$. For each n, $A_n \in \mathcal{B} \subseteq \mathcal{A}_i$, so that $(A_n) \subseteq \mathcal{A}_i$. Since $\mathcal{A}_i$ is closed under countable unions, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}_i$. This holds for every $i \in I$, which implies $\bigcup_n A_n \in \mathcal{B}$. □

Remark. This proposition is what allows defining the *σ-algebra generated* by a family: given a set $\mathcal{F} \subseteq \mathcal{P}(X)$ there is always at least one σ-algebra containing $\mathcal{F}$ (for example $\mathcal{P}(X)$), and the intersection of all of them is the smallest σ-algebra with that property.

2.2 σ-algebra generated

Definition 2.2 – Generated σ-algebra

Given a family $\mathcal{F} \subseteq \mathcal{P}(X)$, the σ-algebra generated by $\mathcal{F}$ is

$$\sigma(\mathcal{F}) = \bigcap \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra and } \mathcal{F} \subseteq \mathcal{A}\}.$$

It is the smallest σ-algebra containing $\mathcal{F}$.

2.3 Structure on countable sets

On a countable set, every σ-algebra comes from a partition, whose elements are called *atoms*: the smallest nonempty measurable sets containing a given point.

Theorem 2.1 – σ-algebras on countable sets

Let X be countable and Σ a σ-algebra on X. For each $x \in X$ define

$$A_x = \bigcap_{\substack{B\in\Sigma \\ x\in B}} B.$$

The family $\mathcal{A} = \{A_x : x \in X\}$ is a partition of X by elements of Σ, and $\Sigma = \sigma(\mathcal{A})$.

Proof. Each B in the intersection contains x, so that $x \in A_x$. In particular the A_x cover X.

To see that $A_x \in \Sigma$, the intersection is reformulated in terms of the complement, which is at most countable. For each $y \in A_x^c$, since y does not belong to the intersection, there is some $B_y \in \Sigma$ with $x \in B_y$ but $y \notin B_y$. Define

$$A'_x = \bigcap_{y \in A_x^c} B_y,$$

which is an at most countable intersection of elements of Σ and therefore belongs to Σ. Each B_y is among the sets that define A_x, which gives $A_x \subseteq B_y$ for every $y \in A_x^c$, and then $A_x \subseteq A'_x$. Conversely, if $y \in A_x^c$ then $y \notin B_y$ and therefore $y \notin A'_x$, which shows $A_x^c \subseteq (A'_x)^c$, that is $A'_x \subseteq A_x$. It follows that $A_x = A'_x \in \Sigma$.

Now a minimality property is shown: if $B \in \Sigma$ is nonempty and $B \subseteq A_x$, then $B = A_x$. Suppose first that $x \notin B$. The complement B^c belongs to Σ and contains x, so that it is among the sets that define A_x; then $A_x \subseteq B^c$, equivalent to $A_x \cap B = \emptyset$. But $B \subseteq A_x$ nonempty forces $A_x \cap B = B \neq \emptyset$, a contradiction. Thus $x \in B$, and since B is a measurable set containing x, $A_x \subseteq B$. Together with the hypothesis $B \subseteq A_x$, one gets $B = A_x$.

The partition property follows from this. If $A_x \cap A_y$ is nonempty, that intersection is in Σ, is nonempty and is contained in A_x, so that minimality forces $A_x \cap A_y = A_x$. Consequently $A_x \subseteq A_y$, and exchanging x with y one obtains $A_y \subseteq A_x$, that is $A_x = A_y$. Combined with the fact that the A_x cover X, this proves that $\mathcal{A}$ is a partition.

It remains to see $\Sigma = \sigma(\mathcal{A})$. The inclusion $\sigma(\mathcal{A}) \subseteq \Sigma$ is a consequence of each $A_x \in \Sigma$. For the other inclusion, given $B \in \Sigma$ it is claimed that $B = \bigcup_{x \in B} A_x$. The inclusion $\subseteq$ follows from $x \in A_x$ for each $x \in B$. For the inclusion $\supseteq$, if $x \in B$ then B is one of the sets that define A_x, which gives $A_x \subseteq B$. The union is countable because $B \subseteq X$ is, and each $A_x \in \sigma(\mathcal{A})$, so that $B \in \sigma(\mathcal{A})$. □

Remark. The σ-algebras on countable sets are then always *discrete*: they are generated by a partition.

Definition 2.3 – Borel σ-algebra

The Borel σ-algebra on $\mathbb{R}$ is the one generated by the open sets:

$$\mathcal{B}(\mathbb{R}) = \sigma(\{U \subseteq \mathbb{R} : U \text{ open}\}).$$

Its elements are called *Borel sets*.

2.4 Borel from intervals

Although the Borel σ-algebra is defined from *all* open sets, on $\mathbb{R}$ a much smaller family suffices: the open intervals (a, b). Something more is proved: every open subset of $\mathbb{R}$ decomposes as an at most countable *disjoint* union of open intervals. These intervals are the connected components of the open set.

Proposition 2.2 – Structure of open sets in $\mathbb{R}$

Every open $A \subseteq \mathbb{R}$ is written as an at most countable union of disjoint open intervals. In particular,

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) : a < b\}).$$

Proof. The strategy is to assign to each point $a \in A$ the largest open interval contained in A that contains it. These intervals cover A and, by their maximality, are pairwise disjoint. Since each one contains a rational, the number of distinct intervals turns out to be at most countable.

For each $a \in A$ define

$$x(a) = \sup\{x \in \mathbb{R} \setminus A : x < a\}, \qquad y(a) = \inf\{y \in \mathbb{R} \setminus A : y > a\},$$

with the convention $\sup \emptyset = -\infty$, $\inf \emptyset = +\infty$, and define $I_a = (x(a), y(a))$.

Since A is open and $a \in A$, there is $\delta > 0$ with $(a - \delta, a + \delta) \subseteq A$. Since no point of the neighborhood $(a - \delta, a)$ is in $\mathbb{R} \setminus A$, every point in the set whose supremum is $x(a)$ satisfies $x \leq a - \delta$. Therefore $x(a) \leq a - \delta < a$, and likewise $y(a) > a$, so that $a \in I_a$.

For the inclusion $I_a \subseteq A$, let $t \in I_a$ with $t \neq a$, and suppose by contradiction $t \notin A$. If $t < a$, then t belongs to the set $\{x \in \mathbb{R} \setminus A : x < a\}$ whose supremum is $x(a)$, which forces $t \leq x(a)$ and contradicts $t > x(a)$. The case $t > a$ is symmetric with respect to $y(a)$.

As a consequence, $A = \bigcup_{a \in A} I_a$: the inclusion $\supseteq$ is $I_a \subseteq A$, and the $\subseteq$ one is $a \in I_a$.

Suppose $I_a \cap I_b \neq \emptyset$ and let $z \in I_a \cap I_b$. By definition of $I_a = (x(a), y(a))$ and $I_b = (x(b), y(b))$, the inequalities

$$x(a) < z < y(a) \qquad \text{and} \qquad x(b) < z < y(b)$$

hold.

First $y(a) = y(b)$ is proved by ruling out the possibility $y(a) < y(b)$ (the case $y(a) > y(b)$ is symmetric, exchanging the roles of a and b). Indeed, suppose $y(a) < y(b)$. By definition of $y(a)$ as the infimum of $\{y \in \mathbb{R} \setminus A : y > a\}$, $y(a) \notin A$. By the inequalities $z < y(a) < y(b)$, one has $y(a) \in (z, y(b))$. But $(z, y(b)) \subseteq I_b \subseteq A$, whence $y(a) \in A$, contradicting $y(a) \notin A$. It follows that $y(a) = y(b)$.

An analogous argument on the other side proves $x(a) = x(b)$: if $x(a) > x(b)$, by definition of $x(a)$ as the supremum of $\{x \in \mathbb{R} \setminus A : x < a\}$, $x(a) \notin A$. By the inequalities $x(b) < x(a) < z$, one has $x(a) \in (x(b), z)$. But $(x(b), z) \subseteq I_b \subseteq A$, whence $x(a) \in A$, contradicting $x(a) \notin A$. The case $x(a) < x(b)$ is symmetric. It follows that $x(a) = x(b)$.

With $x(a) = x(b)$ and $y(a) = y(b)$ it follows that $I_a = I_b$.

Each I_a is a nonempty open subset of $\mathbb{R}$, so that by density of $\mathbb{Q}$ it contains at least one rational. Choosing one rational per distinct interval yields an injective map from the family of distinct intervals into $\mathbb{Q}$ (it is injective because two distinct intervals are disjoint). Consequently, the family injects into $\mathbb{Q}$ and is at most countable.

Let $\mathcal{I} = \{(a, b) : a < b\}$. For one inclusion, each (a, b) is open, which gives $\mathcal{I} \subseteq \mathcal{B}(\mathbb{R})$ and, by minimality, $\sigma(\mathcal{I}) \subseteq \mathcal{B}(\mathbb{R})$. For the other inclusion, every open set is written as a countable union of elements of $\mathcal{I}$ by what was proved, and therefore belongs to $\sigma(\mathcal{I})$. Thus, $\sigma(\mathcal{I})$ is a σ-algebra containing the open sets, and by minimality of $\mathcal{B}(\mathbb{R})$, $\mathcal{B}(\mathbb{R}) \subseteq \sigma(\mathcal{I})$. □

Corollary 2.1 – Sets in $\mathcal{B}(\mathbb{R})$

$\mathcal{B}(\mathbb{R})$ contains the open sets, the closed sets, the points, the half-open intervals and the half-lines.

2.5 Operations with sequences

Definition 2.4 – Limit superior and inferior of sets

Given $(A_n) \subset \mathcal{A}$:

$$\overline{\lim_n} A_n = \bigcap_{m=1}^{\infty} \bigcup_{k \geq m} A_k = \{x : x \in A_n \text{ for infinitely many } n\},$$

$$\underline{\lim_n} A_n = \bigcup_{m=1}^{\infty} \bigcap_{k \geq m} A_k = \{x : x \in A_n \text{ from some } n \text{ on}\}.$$

Remark. If $\mathcal{A}$ is a σ-algebra and $(A_n) \subset \mathcal{A}$, then $\overline{\lim} A_n, \underline{\lim} A_n \in \mathcal{A}$.

Proposition 2.3 – Properties of the limit superior and inferior

Let (A_n) be a sequence of subsets of X. Then:
(i) $\underline{\lim}_n A_n \subseteq \overline{\lim}_n A_n$.
(ii) $(\underline{\lim}_n A_n)^c = \overline{\lim}_n A_n^c$ and $\left(\overline{\lim}_n A_n\right)^c = \underline{\lim}_n A_n^c$.
(iii) If $A \subseteq A_n \subseteq B$ for every n, then $A \subseteq \underline{\lim}_n A_n \subseteq \overline{\lim}_n A_n \subseteq B$.
(iv) If (A_n) is increasing, $\underline{\lim}_n A_n = \overline{\lim}_n A_n = \bigcup_n A_n$.
(v) If (A_n) is decreasing, $\underline{\lim}_n A_n = \overline{\lim}_n A_n = \bigcap_n A_n$.

Proof.
(i) If $x \in \underline{\lim}_n A_n = \bigcup_m \bigcap_{k \geq m} A_k$, there is m_0 such that $x \in A_k$ for every $k \geq m_0$; that is, x is in all the A_k except finitely many. In particular, x is in infinitely many A_k, so that for every m there is $k \geq m$ with $x \in A_k$. This means $x \in \bigcap_m \bigcup_{k \geq m} A_k = \overline{\lim}_n A_n$.

(ii) By De Morgan's laws applied twice,

$$\left(\underline{\lim_n} A_n\right)^c = \left(\bigcup_m \bigcap_{k \geq m} A_k\right)^c = \bigcap_m \left(\bigcap_{k \geq m} A_k\right)^c = \bigcap_m \bigcup_{k \geq m} A_k^c = \overline{\lim_n} A_n^c.$$

The other equality is proved analogously, exchanging $\bigcup$ and $\bigcap$.

(iii) If $x \in A$, then $x \in A_n$ for every n, in particular $x \in \bigcap_{k \geq m} A_k$ for every m. Hence $x \in \bigcup_m \bigcap_{k \geq m} A_k = \underline{\lim}_n A_n$, which gives $A \subseteq \underline{\lim}_n A_n$. Combined with (i), $A \subseteq \underline{\lim}_n A_n \subseteq \overline{\lim}_n A_n$.

For the upper bound, $A_n \subseteq B$ for every n is equivalent to $B^c \subseteq A_n^c$ for every n. Applying the inclusion $B^c \subseteq \underline{\lim}_n A_n^c$ (just proved with B^c in place of A and A_n^c in place of A_n) and then (ii),

$$B^c \subseteq \underline{\lim_n} A_n^c = \left(\overline{\lim_n} A_n\right)^c,$$

whence, taking complements, $\overline{\lim}_n A_n \subseteq B$.

(iv) If (A_n) is increasing, $\bigcap_{k \geq m} A_k = A_m$ for every m, so that

$$\underline{\lim_n} A_n = \bigcup_m \bigcap_{k \geq m} A_k = \bigcup_m A_m.$$

On the other hand, $\bigcup_{k\ge m} A_k = \bigcup_k A_k$ for every m (the union does not change when starting later in an increasing sequence), so that

$$\overline{\lim_n}\, A_n = \bigcap_m \bigcup_{k\ge m} A_k = \bigcup_n A_n.$$

(v) If (A_n) is decreasing, (A_n^c) is increasing. Applying (iv) to (A_n^c) and then (ii),

$$\left(\overline{\lim_n}\, A_n\right)^c = \underline{\lim_n}\, A_n^c = \bigcup_n A_n^c = \left(\bigcap_n A_n\right)^c,$$

and analogously for $\underline{\lim}$. Taking complements, $\underline{\lim}_n A_n = \overline{\lim}_n A_n = \bigcap_n A_n$. □

Example 2.5 – Computing limits of sets

(i) Let $A_n = [0,2]$ if n is odd and $A_n = [1,4]$ if n is even. A point is in infinitely many A_n if and only if it belongs to $[0,2]\cup[1,4] = [0,4]$, and it is in all but finitely many if and only if it belongs to $[0,2]\cap[1,4] = [1,2]$. Thus

$$\underline{\lim_n}\, A_n = [1,2], \qquad \overline{\lim_n}\, A_n = [0,4],$$

and the sequence does not converge as sets.

(ii) Let $A_n = [1/n, 1)$. The sequence is increasing, and by Proposition 2.3,

$$\underline{\lim_n}\, A_n = \overline{\lim_n}\, A_n = \bigcup_n [1/n, 1) = (0,1).$$

The sequence converges to the set $(0,1)$; observe that the limit does not contain 0 even though each A_n is "attached" to it.

Remark. In the space $X = \{0,1\}^{\mathbb{N}}$ of coin tosses with the events $E_n = \{x : x_n = 1\}$, the limits have a direct probabilistic reading:

$$\overline{\lim_n}\, E_n = \{\text{heads infinitely often}\}, \qquad \underline{\lim_n}\, E_n = \{\text{eventually always heads}\}.$$

The inclusion $\underline{\lim} \subseteq \overline{\lim}$ is, with this reading, transparent: if from some moment on heads always comes up, in particular heads comes up infinitely often.

3 Non-measurable sets: Vitali's example

Before moving on to the definition of measure and to Lebesgue's construction, it is convenient to understand why one cannot expect *every* subset of $\mathbb{R}$ to admit a measure with the desired properties (translation invariance and σ-additivity). The classical example showing that this is impossible is *Vitali's* construction: a subset $V \subseteq [0,1]$ that cannot be measurable with respect to any translation-invariant measure extending the "length" $m([0,1]) = 1$.

Theorem 3.1 – Existence of non-measurable sets

There is no function $m : \mathcal{P}(\mathbb{R}) \to [0,\infty]$ that simultaneously satisfies:
(i) m is σ-additive: if (A_n) is disjoint, $m(\bigcup_n A_n) = \sum_n m(A_n)$.
(ii) m is translation invariant: $m(A+t) = m(A)$ for every $A \subseteq \mathbb{R}$ and every $t \in \mathbb{R}$.
(iii) $m([0,1]) = 1$.

Proof.
Assuming the axiom of choice, a $V \subseteq [0,1]$ is constructed that will lead to a contradiction under (i)–(iii).

Construction. On $[0,1]$ define the equivalence relation

$$x \sim y \;\Leftrightarrow\; x - y \in \mathbb{Q}.$$

Each class $[x]$ is dense in $[0,1]$ (because $\mathbb{Q}$ is dense). By the axiom of choice, choose one representative for each class and form

$$V = \{\text{one representative of each class}\} \subseteq [0,1].$$

Translates. Enumerate $\mathbb{Q} \cap [-1,1] = \{r_1, r_2, \dots\}$, which is countable. For each k define

$$V_k = V + r_k = \{v + r_k : v \in V\}.$$

The V_k are pairwise disjoint. If $V_k \cap V_j \neq \emptyset$, there are $v_1, v_2 \in V$ with $v_1 + r_k = v_2 + r_j$, whence $v_1 - v_2 = r_j - r_k \in \mathbb{Q}$, that is $v_1 \sim v_2$. Since V contains a single representative per class, $v_1 = v_2$, and then $r_k = r_j$, that is $k = j$.

$[0,1] \subseteq \bigcup_k V_k \subseteq [-1,2]$. The inclusion on the right is a consequence of the fact that $V \subseteq [0,1]$ and $r_k \in [-1,1]$ give $V_k \subseteq [-1,2]$. For the inclusion on the left, given $x \in [0,1]$, its class $[x]$ has a representative $v \in V$, and $x - v \in \mathbb{Q} \cap [-1,1]$ (because $x, v \in [0,1]$), so that $x - v = r_k$ for some k, that is $x = v + r_k \in V_k$.

Contradiction. Suppose that V is measurable. By (ii), $m(V_k) = m(V)$ for every k, and by (i),

$$m\left(\bigcup_k V_k\right) = \sum_{k=1}^{\infty} m(V_k) = \sum_{k=1}^{\infty} m(V).$$

Cases are distinguished according to the value of $m(V) \in [0,\infty]$:

- If $m(V) = 0$, the sum gives 0, and by monotonicity $m([0,1]) \leq m(\bigcup_k V_k) = 0$. But $m([0,1]) = 1$ by (iii), a contradiction.
- If $m(V) > 0$, $\sum_k m(V) = +\infty$, whereas by monotonicity $m(\bigcup_k V_k) \leq m([-1,2]) = 3$ (the latter by translation of $[0,1]$ and additivity: $m([-1,2]) = m([-1,0]) + m([0,1]) + m([1,2]) = 3$). Contradiction.

In either case a contradiction is reached. Therefore V cannot be measurable with respect to m, and in particular there is no function $m : \mathcal{P}(\mathbb{R}) \to [0, \infty]$ satisfying (i)–(iii) on all of $\mathcal{P}(\mathbb{R})$. $\square$

Remark. A translation-invariant, σ-additive and normalized measure cannot be defined on *the whole* σ-algebra $\mathcal{P}(\mathbb{R})$. The most that can be achieved (with the axiom of choice) is a measure on a strictly smaller σ-algebra: the Borel sets, or somewhat more: the Lebesgue-measurable sets.

4 Measures

Definition 4.1 – Measure

Let $(X, \mathcal{A})$ be a pair with $\mathcal{A}$ a σ-algebra on X. A function $\mu : \mathcal{A} \to [0, \infty]$ is a *measure* if:
(i) $\mu(\emptyset) = 0$.
(ii) (σ-additivity) If $(A_n) \subset \mathcal{A}$ is a pairwise disjoint family,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The triple $(X, \mathcal{A}, \mu)$ is called a *measure space*.
If $\mu(X) < \infty$, μ is said to be *finite*. If $X = \bigcup_n X_n$ with $\mu(X_n) < \infty$ for every n, μ is said to be *σ-finite*. If $\mu(X) = 1$, μ is a *probability*.

4.1 Examples

Proposition 4.1 – Dirac delta

Let X be a nonempty set and $x_0 \in X$ a fixed point. The function $\delta_{x_0} : \mathcal{P}(X) \to [0, \infty]$ defined by

$$\delta_{x_0}(A) = \begin{cases} 1, & x_0 \in A, \\ 0, & x_0 \notin A, \end{cases}$$

is a measure (called the *Dirac delta* concentrated at x_0).

Proof.
$\delta_{x_0}(\emptyset) = 0$. Since $x_0 \notin \emptyset$, $\delta_{x_0}(\emptyset) = 0$.

σ-additivity. Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{P}(X)$ be a pairwise disjoint family, and $A = \bigcup_n A_n$. Two cases are distinguished according to whether x_0 is in A or not.

Case 1: $x_0 \in A$. There is $n_0 \in \mathbb{N}$ with $x_0 \in A_{n_0}$. Since the A_n are pairwise disjoint, that n_0 is unique: for every $n \neq n_0$, $x_0 \notin A_n$. Therefore

$$\sum_{n=1}^{\infty} \delta_{x_0}(A_n) = \delta_{x_0}(A_{n_0}) + \sum_{n\neq n_0} \delta_{x_0}(A_n) = 1 + 0 = 1 = \delta_{x_0}(A).$$

Case 2: $x_0 \notin A$. Then $x_0 \notin A_n$ for every n, so that $\delta_{x_0}(A_n) = 0$ for every n, and

$$\sum_{n=1}^{\infty} \delta_{x_0}(A_n) = 0 = \delta_{x_0}(A).$$

□

Remark. δ_{x_0} is a probability ($\delta_{x_0}(X) = 1$). It concentrates all the mass at the point x_0.

Proposition 4.2 – Counting measure

Let X be a nonempty set. The function $\mu : \mathcal{P}(X) \to [0, \infty]$ defined by

$$\mu(A) = \#A = \begin{cases} \text{number of elements of } A, & \text{if } A \text{ is finite,} \\ \infty, & \text{if } A \text{ is infinite,} \end{cases}$$

is a measure (called the *counting measure*).

Proof.
$\mu(\emptyset) = 0$. The empty set has zero elements.

σ-additivity. Let $(A_n)_{n\in\mathbb{N}}$ be pairwise disjoint and $A = \bigcup_n A_n$. The structure of the proof depends on whether A is finite or infinite.

Case 1: A finite. Since each $A_n \subseteq A$, all the A_n are finite. Moreover, since the A_n are disjoint and A is finite, only finitely many of them can be nonempty; otherwise A would contain at least one element for each nonempty A_n, contradicting $\#A < \infty$. Renumbering, suppose $A_1, \dots, A_N$ nonempty and $A_n = \emptyset$ for $n > N$. By elementary counting of finite disjoint unions,

$$\#A = \#(A_1 \sqcup \cdots \sqcup A_N) = \sum_{n=1}^{N} \#A_n = \sum_{n=1}^{\infty} \#A_n,$$

where the last equality uses that $\#A_n = 0$ for $n > N$.

Case 2: A infinite. Then $\mu(A) = \infty$. Two subcases are distinguished:

- If there is n_0 with A_{n_0} infinite, then $\#A_{n_0} = \infty$, and
$$\sum_{n=1}^{\infty} \#A_n \geq \#A_{n_0} = \infty.$$
- If all the A_n are finite, since A is infinite there must be infinitely many nonempty A_n; otherwise A would be a finite union of finite sets, and therefore finite. The series of positive terms
$$\sum_{n=1}^{\infty} \#A_n$$
then has infinitely many terms ≥ 1, and diverges to ∞.

In both subcases $\sum_n \#A_n = \infty = \#A$. □

Remark. The counting measure is finite if and only if X is finite, and it is σ-finite if and only if X is at most countable. On $X = \mathbb{N}$ with the counting measure, integrals with respect to μ reduce to series.

Example 4.1 – Bernoulli measure

On $X = \{0,1\}^{\mathbb{N}}$ with the σ-algebra generated by the events $E_n = \{x : x_n = 1\}$ (Example 2.4), there is a unique probability P that models independent tosses of a fair coin: on the events that fix k coordinates,

$$P(\{x : x_{i_1} = \epsilon_1, \dots, x_{i_k} = \epsilon_k\}) = \frac{1}{2^k}.$$

With it the probability of *heads at some point* can be computed. The event is $\bigcup_n E_n$, which is not a disjoint union; it is made disjoint with

$$\begin{aligned} A_n = E_n \setminus (E_1 \cup \cdots \cup E_{n-1}) &= \{x : x_1 = \cdots = x_{n-1} = 0,\ x_n = 1\} \\ &= \{\text{the first heads is the } n\text{-th toss}\}, \end{aligned}$$

which fixes n coordinates, so that $P(A_n) = 1/2^n$. By σ-additivity,

$$P\left(\bigcup_n E_n\right) = \sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1.$$

With probability 1 some heads comes up.

Remark. Of the previous examples, only the Dirac delta and the counting measure have been *constructed*. For the Bernoulli measure, and for the Lebesgue measure on $\mathbb{R}$ (which assigns to each interval its length), existence has not been proved so far: that is the goal of the section on generating measures.

4.2 Basic properties

Theorem 4.1 – Properties of a measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. For $A, B, A_n \in \mathcal{A}$:

(i) If $E \subseteq F$, $\mu(E) \leq \mu(F)$.

(ii) If $E \subseteq F$ and $\mu(E) < \infty$, $\mu(F \setminus E) = \mu(F) - \mu(E)$.

(iii) If $\mu(A \cap B) < \infty$,

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

(iv) For any (not necessarily disjoint) family (A_n),

$$\mu\left(\bigcup_n A_n\right) \leq \sum_n \mu(A_n).$$

(v) If $A_n \nearrow A$ (that is, $A_n \subseteq A_{n+1}$ and $A = \bigcup_n A_n$),

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

(vi) If $A_n \searrow A$ and $\mu(A_1) < \infty$,

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

(vii) If $\overline{\lim}_n A_n = \underline{\lim}_n A_n = A$ (that is, (A_n) converges to A as sets) and $\mu(\bigcup_n A_n) < \infty$,

$$\mu(A) = \lim_{n\to\infty} \mu(A_n).$$

Proof.

(i) $F = E \cup (F \setminus E)$ is a disjoint union, so that $\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$.

(ii) It follows from the previous step by solving for $\mu(F \setminus E)$.

(iii) Decompose $A \cup B$ into three disjoint sets: $A \cup B = (A \setminus B) \sqcup (B \setminus A) \sqcup (A \cap B)$. By σ-additivity,

$$\mu(A \cup B) = \mu(A \setminus B) + \mu(B \setminus A) + \mu(A \cap B).$$

Analogously, $A = (A \setminus B) \sqcup (A \cap B)$ and $B = (B \setminus A) \sqcup (A \cap B)$, whence

$$\mu(A) + \mu(B) = \mu(A \setminus B) + \mu(B \setminus A) + 2\mu(A \cap B).$$

Subtracting $\mu(A \cap B)$ (finite by hypothesis, which avoids indeterminate forms $\infty - \infty$) yields $\mu(A) + \mu(B) - \mu(A \cap B) = \mu(A \cup B)$.

(iv) Make the family disjoint: $B_1 = A_1$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. The B_n are disjoint, $B_n \subseteq A_n$ (so $\mu(B_n) \leq \mu(A_n)$), and $\bigcup B_n = \bigcup A_n$. Hence

$$\mu\Big(\bigcup A_n\Big) = \sum \mu(B_n) \leq \sum \mu(A_n).$$

(v) If $A_n \nearrow A$, make it disjoint with $B_1 = A_1$, $B_n = A_n \setminus A_{n-1}$. Then $A_n = \bigsqcup_{k=1}^{n} B_k$ and $A = \bigsqcup_k B_k$, so that

$$\mu(A) = \sum_{k=1}^{\infty} \mu(B_k) = \lim_n \sum_{k=1}^{n} \mu(B_k) = \lim_n \mu(A_n).$$

(vi) If $A_n \searrow A$, define $C_n = A_1 \setminus A_n$, which increases to $A_1 \setminus A$. Applying (v) and (ii) (using $\mu(A_1) < \infty$),

$$\mu(A_1) - \mu(A) = \mu(A_1 \setminus A) = \lim_n \mu(C_n) = \lim_n (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \lim_n \mu(A_n).$$

Cancelling $\mu(A_1)$, $\mu(A) = \lim \mu(A_n)$.

(vii) Define

$$B_m = \bigcap_{k \geq m} A_k, \qquad C_m = \bigcup_{k \geq m} A_k.$$

By construction, $B_m \nearrow \underline{\lim}_n A_n = A$ and $C_m \searrow \overline{\lim}_n A_n = A$. Moreover, $B_m \subseteq A_m \subseteq C_m$ by monotonicity of intersection and union.

By (v) applied to (B_m),

$$\lim_m \mu(B_m) = \mu(A).$$

By (vi) applied to (C_m) (which is legitimate because $\mu(C_1) \leq \mu(\bigcup_n A_n) < \infty$),

$$\lim_m \mu(C_m) = \mu(A).$$

By (i), $\mu(B_m) \leq \mu(A_m) \leq \mu(C_m)$ for every m. By the squeeze theorem, $\lim_m \mu(A_m) = \mu(A)$. □

4.3 Borel-Cantelli lemmas

Recall the probabilistic interpretation of $\overline{\lim} A_n$: a point ω belongs to $\overline{\lim} A_n = \bigcap_m \bigcup_{k \geq m} A_k$ if and only if it is in infinitely many A_n. That is why one usually writes $\{A_n \text{ infinitely often}\}$. The Borel-Cantelli lemmas give conditions under which this event has measure zero or one.

Theorem 4.2 – Borel-Cantelli (I)

Let $(X, \mathcal{A}, \mu)$ be a measure space and $(A_n) \subset \mathcal{A}$. If

$$\sum_{n=1}^{\infty} \mu(A_n) < \infty,$$

then $\mu(\overline{\lim}_n A_n) = 0$.

Proof. By definition, $\overline{\lim}_n A_n = \bigcap_{m=1}^{\infty} C_m$, where $C_m = \bigcup_{k \geq m} A_k$. By sub-$\sigma$-additivity,

$$\mu(C_m) = \mu\left(\bigcup_{k \geq m} A_k\right) \leq \sum_{k=m}^{\infty} \mu(A_k).$$

Since $\sum_n \mu(A_n) < \infty$, its tail $\sum_{k \geq m} \mu(A_k)$ tends to 0 as $m \to \infty$. In particular, $\mu(C_1) \leq \sum_{k=1}^{\infty} \mu(A_k) < \infty$, so that the decreasing sequence (C_m) satisfies the hypotheses of continuity along decreasing sequences. Then

$$\mu\left(\overline{\lim_n} A_n\right) = \mu\left(\bigcap_m C_m\right) = \lim_{m \to \infty} \mu(C_m) \leq \lim_{m \to \infty} \sum_{k=m}^{\infty} \mu(A_k) = 0.$$

By monotonicity, $\mu(\overline{\lim} A_n) = 0$. □

Remark. Borel-Cantelli (I) requires neither independence of the A_n nor that μ be a probability: it holds for every measure.

Definition 4.2 – Independence of events

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

(i) Two events $A, B \in \mathcal{F}$ are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

(ii) A finite family $A_1, \dots, A_n \in \mathcal{F}$ is *independent* if for every subset of indices $\{i_1, \dots, i_k\} \subseteq \{1, \dots, n\}$ (with $k \geq 1$),

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}).$$

(iii) A family $(A_n)_{n \in \mathbb{N}}$ is *independent* if every finite subfamily of it is.

Remark. If the (A_n) are independent, so are the complements (A_n^c) (in any finite subfamily). This follows by inclusion-exclusion: for example, $\mathbb{P}(A_1^c \cap A_2^c) = 1 - \mathbb{P}(A_1) - \mathbb{P}(A_2) + \mathbb{P}(A_1)\mathbb{P}(A_2) = (1 - \mathbb{P}(A_1))(1 - \mathbb{P}(A_2)) = \mathbb{P}(A_1^c)\mathbb{P}(A_2^c)$; the general case by induction.

Theorem 4.3 – Borel-Cantelli (II)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(A_n) \subset \mathcal{F}$ an *independent* sequence. If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then $\mathbb{P}(\overline{\lim}_n A_n) = 1$.

Proof. It will suffice to prove $\mathbb{P}((\overline{\lim}_n A_n)^c) = 0$. By the properties seen earlier,

$$(\overline{\lim_n} A_n)^c = \underline{\lim_n} A_n^c.$$

By sub-σ-additivity, it is enough to see that for every m,

$$\mathbb{P}\left(\bigcap_{k \geq m} A_k^c \right) = 0.$$

Fix m. By continuity along decreasing sequences (the finite intersection $\bigcap_{k=m}^{M} A_k^c$ decreases to $\bigcap_{k \geq m} A_k^c$ and has probability $\leq 1 < \infty$),

$$\mathbb{P}\left(\bigcap_{k \geq m} A_k^c \right) = \lim_{M \to \infty} \mathbb{P}\left(\bigcap_{k=m}^{M} A_k^c \right).$$

Since (A_n) is independent, so is (A_n^c). Applying the factorization to the finite family $A_m^c, \dots, A_M^c$,

$$\mathbb{P}\left(\bigcap_{k=m}^{M} A_k^c \right) = \prod_{k=m}^{M} \mathbb{P}(A_k^c) = \prod_{k=m}^{M} (1 - \mathbb{P}(A_k)).$$

The inequality $1 - x \leq e^{-x}$, valid for every $x \in \mathbb{R}$, is used:

$$\prod_{k=m}^{M} (1 - \mathbb{P}(A_k)) \leq \prod_{k=m}^{M} e^{-\mathbb{P}(A_k)} = \exp\left(- \sum_{k=m}^{M} \mathbb{P}(A_k) \right).$$

Since $\sum_n \mathbb{P}(A_n) = \infty$, the tail $\sum_{k=m}^{M} \mathbb{P}(A_k) \to \infty$ as $M \to \infty$ (for each fixed m). Therefore the exponent tends to $-\infty$ and

$$\lim_{M\to\infty} \mathbb{P}\left(\bigcap_{k=m}^{M} A_k^c\right) \le \lim_{M\to\infty} \exp\left(-\sum_{k=m}^{M} \mathbb{P}(A_k)\right) = 0,$$

that is, $\mathbb{P}(\bigcap_{k\ge m} A_k^c) = 0$. Since this holds for every m,

$$\mathbb{P}\big((\overline{\lim}\, A_n)^c\big) = \mathbb{P}\left(\bigcup_{m}\bigcap_{k\ge m} A_k^c\right) \le \sum_{m} \mathbb{P}\left(\bigcap_{k\ge m} A_k^c\right) = 0.$$

It follows that $\mathbb{P}(\overline{\lim}\, A_n) = 1$. □

Example 4.2 – Increasingly biased coins

Independent coins are tossed, where the n-th comes up heads with probability p_n, and let E_n = "the n-th comes up heads".

(i) If $p_n = 1/n$: since $\sum 1/n = \infty$ and the events are independent, Borel-Cantelli (II) gives that with probability 1 *infinitely many* heads come up, even though $p_n \to 0$.

(ii) If $p_n = (99/100)^n$: the series $\sum (99/100)^n$ is a convergent geometric series, and Borel-Cantelli (I) gives that with probability 1 only *finitely many* heads come up, even though the first coins come up heads almost always.

Example 4.3 – The monkey and Borges

A monkey types at random, with equiprobable and independent keys, on a keyboard of 60 symbols. Fix a work of N characters (say, a short story by Borges). The claim is: with probability 1, the monkey types it, and not once but *infinitely many* times.

Consider the disjoint blocks of N consecutive keystrokes (the k-th block occupies the positions $(k-1)N+1, \dots, kN$) and let S_k be the event "the k-th block coincides character by character with the work". The S_k depend on disjoint keystrokes, so that they are independent, and each one has probability $p = (1/60)^N$: tiny, but positive and the same for every k. Then

$$\sum_{k=1}^{\infty} \mathbb{P}(S_k) = \sum_{k=1}^{\infty} \frac{1}{60^N} = \infty,$$

and Borel-Cantelli (II) gives $\mathbb{P}(\overline{\lim}_k\, S_k) = 1$: infinitely many blocks reproduce the complete work.

Remark. The example illustrates the typical use of Borel-Cantelli (II): to conclude "infinitely often" the events need not be likely; it suffices that they be independent and that the series of their probabilities diverge, which a constant positive probability guarantees.

Remark. Combining both directions: if the (A_n) are *independent*, then

$$\mathbb{P}(\overline{\lim}\, A_n) = \begin{cases} 0, & \text{if } \sum \mathbb{P}(A_n) < \infty, \\ 1, & \text{if } \sum \mathbb{P}(A_n) = \infty. \end{cases}$$

That is, under independence the event $\{A_n \text{ infinitely often}\}$ always has probability 0 or 1. This dichotomy is a particular case of a much more general phenomenon: Kolmogorov's 0-1 law.

4.4 Kolmogorov's 0-1 law

Definition 4.3 – Tail σ-algebra

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(A_n)_{n\in\mathbb{N}} \subset \mathcal{F}$. The *tail σ-algebra* of the sequence (A_n) is

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(A_n, A_{n+1}, A_{n+2}, \dots),$$

where $\sigma(A_n, A_{n+1}, \dots)$ is the σ-algebra generated by the sequence from index n on. Its elements are called *tail events*: they are the ones that do not depend on any finite number of the A_k.

Remark. Examples of tail events: $\overline{\lim}_n A_n$ and $\underline{\lim}_n A_n$ (both are in $\sigma(A_n, A_{n+1}, \dots)$ for every n, by their very definition). The event "the sequence $\mathbf{1}_{A_n}$ converges" is also a tail event. On the other hand, A_1 by itself is not.

Theorem 4.4 – Kolmogorov's 0-1 law

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(A_n)_{n\in\mathbb{N}} \subset \mathcal{F}$ an *independent* sequence. If $B \in \mathcal{T}$ is a tail event, then

$$\mathbb{P}(B) \in \{0, 1\}.$$

Proof. The strategy is to prove that B is independent of itself, which forces $\mathbb{P}(B) = \mathbb{P}(B \cap B) = \mathbb{P}(B)^2$ and therefore $\mathbb{P}(B) \in \{0, 1\}$.

The following criterion for independence of generated σ-algebras is used; it is a consequence of Dynkin's π-λ theorem, and its proof is omitted: *if two families of events $\mathcal{C}_1, \mathcal{C}_2$ are closed under finite intersections and every pair $C_1 \in \mathcal{C}_1$, $C_2 \in \mathcal{C}_2$ is independent, then $\sigma(\mathcal{C}_1)$ and $\sigma(\mathcal{C}_2)$ are independent* (every pair of elements, one from each, is).

$\mathcal{T}$ is independent of $\sigma(A_1, \dots, A_n)$, for each n. Let $\mathcal{C}_1$ be the family of finite intersections of sets A_i with $i \le n$, and $\mathcal{C}_2$ that of finite intersections of sets A_j with $j > n$. Both are closed under finite intersections, and every pair of elements from one and the other is independent because the sequence (A_n) is (the indices involved are all distinct). By the criterion, $\sigma(A_1, \dots, A_n)$ and $\sigma(A_{n+1}, A_{n+2}, \dots)$ are independent; since $\mathcal{T} \subseteq \sigma(A_{n+1}, A_{n+2}, \dots)$, in particular $\mathcal{T}$ is independent of $\sigma(A_1, \dots, A_n)$.

$\mathcal{T}$ is independent of $\sigma(A_1, A_2, \dots)$. Let $\mathcal{C} = \bigcup_n \sigma(A_1, \dots, A_n)$. This family is closed under finite intersections (two of its elements coexist in some $\sigma(A_1, \dots, A_n)$ with n sufficiently large), each of its elements is independent of each element of $\mathcal{T}$ by the previous step, and $\sigma(\mathcal{C}) = \sigma(A_1, A_2, \dots)$. Applying the criterion again (with $\mathcal{C}$ and $\mathcal{T}$, which is closed under intersections because it is a σ-algebra), the independence of $\sigma(A_1, A_2, \dots)$ and $\mathcal{T}$ is obtained.

Conclusion. Let $B \in \mathcal{T}$. Also $B \in \sigma(A_1, A_2, \dots)$, since $\mathcal{T} \subseteq \sigma(A_1, A_2, \dots)$. The independence just proved, applied to the pair (B, B), gives

$$\mathbb{P}(B) = \mathbb{P}(B \cap B) = \mathbb{P}(B) \cdot \mathbb{P}(B),$$

whence $\mathbb{P}(B)(1 - \mathbb{P}(B)) = 0$ and $\mathbb{P}(B) \in \{0, 1\}$. □

Remark. The law does not say *which* of the two values $\mathbb{P}(B)$ takes, and deciding it can be difficult. For the tail event $\overline{\lim}_n A_n$, the Borel-Cantelli lemmas are the effective criterion.

5 Generating measures

5.1 Outer measure

Definition 5.1 – Outer measure

A function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ is an *outer measure* if:
(i) $\mu^*(\emptyset) = 0$.
(ii) Monotonicity: $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$.
(iii) Sub-σ-additivity: $\mu^*(\bigcup_n A_n) \leq \sum_n \mu^*(A_n)$ for any $(A_n) \subseteq \mathcal{P}(X)$.

Remark. *Typical construction of an outer measure.* Given a function $\ell : \mathcal{C} \to [0, \infty]$ defined on a family $\mathcal{C} \subseteq \mathcal{P}(X)$ with $\emptyset \in \mathcal{C}$ and $\ell(\emptyset) = 0$, define

$$\mu^*(E) = \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : (I_n) \subset \mathcal{C},\ E \subseteq \bigcup_n I_n \right\},$$

with $\mu^*(E) = +\infty$ if E admits no cover. Then μ^* is an outer measure .

5.2 Carathéodory's criterion

Definition 5.2 – μ^*-measurable sets

$A \subseteq X$ is *μ^*-measurable* (in the sense of Carathéodory) if for every $E \subseteq X$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Theorem 5.1 – Carathéodory

Let μ^* be an outer measure on X, and let

$$\mathcal{A}^* = \{A \subseteq X : A \text{ is } \mu^*\text{-measurable}\}.$$

Then $\mathcal{A}^*$ is a σ-algebra and $\mu = \mu^*|_{\mathcal{A}^*}$ is a measure.

Proof.
For every $A \subseteq X$ and every $E \subseteq X$, subadditivity gives $\mu^*(E) = \mu^*((E \cap A) \cup (E \cap A^c)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. Therefore, to see that a set is in $\mathcal{A}^*$ it suffices to prove the opposite inequality.

Step 1: $\mathcal{A}^*$ is an algebra. The empty set is in $\mathcal{A}^*$, since $\mu^*(E \cap \emptyset) + \mu^*(E \cap X) = 0 + \mu^*(E) = \mu^*(E)$ for every E. Closure under complements follows from the fact that Carathéodory's condition is symmetric in A and A^c. For closure under finite unions, let $A, B \in \mathcal{A}^*$ and $E \subseteq X$. Applying the measurability of A to E, and then that of B to $E \cap A^c$,

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c). \end{aligned}$$

Since $E \cap (A \cup B) = (E \cap A) \sqcup (E \cap A^c \cap B)$, subadditivity gives $\mu^*(E \cap (A \cup B)) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c \cap B)$. Substituting in the previous equality and using $E \cap (A \cup B)^c = E \cap A^c \cap B^c$,

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c),$$

so that $A \cup B \in \mathcal{A}^*$.

Step 2: $\mathcal{A}^*$ is a σ-algebra. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}^*$ be a disjoint sequence and $A = \bigcup_n A_n$. Write $S_n = \bigcup_{k=1}^n A_k$. It is proved by induction on n that, for every E,

$$\mu^*(E \cap S_n) = \sum_{k=1}^n \mu^*(E \cap A_k).$$

The case $n = 1$ holds because $S_1 = A_1$. Assuming the formula for n, observe that $S_n \in \mathcal{A}^*$ since it is a finite union, and apply Carathéodory's condition for A_{n+1} to the set $E \cap S_{n+1}$:

$$\mu^*(E \cap S_{n+1}) = \mu^*(E \cap S_{n+1} \cap A_{n+1}) + \mu^*(E \cap S_{n+1} \cap A_{n+1}^c).$$

Since the A_k are disjoint, $S_{n+1} \cap A_{n+1} = A_{n+1}$ and $S_{n+1} \cap A_{n+1}^c = S_n$, so that

$$\mu^*(E \cap S_{n+1}) = \mu^*(E \cap A_{n+1}) + \mu^*(E \cap S_n).$$

Applying the inductive hypothesis to $\mu^*(E \cap S_n)$,

$$\mu^*(E \cap S_{n+1}) = \mu^*(E \cap A_{n+1}) + \sum_{k=1}^n \mu^*(E \cap A_k) = \sum_{k=1}^{n+1} \mu^*(E \cap A_k),$$

which is the formula for $n + 1$.

Now, by the measurability of S_n,

$$\mu^*(E) = \mu^*(E \cap S_n) + \mu^*(E \cap S_n^c).$$

In the first term the formula just proved is used; in the second, since $S_n \subseteq A$ gives $S_n^c \supseteq A^c$, monotonicity ensures $\mu^*(E \cap S_n^c) \geq \mu^*(E \cap A^c)$. Therefore

$$\mu^*(E) \geq \sum_{k=1}^n \mu^*(E \cap A_k) + \mu^*(E \cap A^c),$$

and since this holds for every n, letting $n \to \infty$,

$$\mu^*(E) \geq \sum_{k=1}^\infty \mu^*(E \cap A_k) + \mu^*(E \cap A^c).$$

Finally, $E \cap A = \bigcup_k (E \cap A_k)$, so that sub-$\sigma$-additivity gives $\sum_{k=1}^\infty \mu^*(E \cap A_k) \geq \mu^*(E \cap A)$, and then

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c),$$

that is, $A \in \mathcal{A}^*$. For an arbitrary countable union $(B_n) \subseteq \mathcal{A}^*$, make it disjoint: $A_1 = B_1$ and $A_n = B_n \setminus (B_1 \cup \cdots \cup B_{n-1})$ for $n \geq 2$. Since $\mathcal{A}^*$ is an algebra, each $A_n \in \mathcal{A}^*$, the A_n are disjoint and $\bigcup_n B_n = \bigcup_n A_n \in \mathcal{A}^*$ by the above. Hence $\mathcal{A}^*$ is a σ-algebra.

Step 3: $\mu = \mu^*|_{\mathcal{A}^*}$ is a measure. Let $(A_n) \subseteq \mathcal{A}^*$ be disjoint and $A = \bigcup_n A_n$. Evaluate the inequality obtained in Step 2 at $E = A$. Since $A \cap A_k = A_k$ and $A \cap A^c = \emptyset$,

$$\mu^*(A) \geq \sum_{k=1}^\infty \mu^*(A \cap A_k) + \mu^*(A \cap A^c) = \sum_{k=1}^\infty \mu^*(A_k).$$

The opposite inequality $\mu^*(A) \leq \sum_k \mu^*(A_k)$ is sub-σ-additivity, so that $\mu(A) = \sum_n \mu(A_n)$; that is, μ is σ-additive on disjoint sequences of $\mathcal{A}^*$. Since moreover $\mu(\emptyset) = \mu^*(\emptyset) = 0$, μ is a measure. □

5.3 Lebesgue measure on $\mathbb{R}$

Definition 5.3 – Lebesgue outer measure

For $E \subseteq \mathbb{R}$, define

$$\lambda^*(E) = \inf\left\{\sum_{n=1}^{\infty}(b_n - a_n) : E \subseteq \bigcup_n (a_n, b_n]\right\},$$

Theorem 5.2 – Lebesgue measure

λ^* is an outer measure. The associated Carathéodory σ-algebra, denoted $\mathcal{L}$, contains $\mathcal{B}(\mathbb{R})$, and the restriction $\lambda = \lambda^*|_{\mathcal{L}}$ is a measure (called *Lebesgue measure*) that satisfies:

(i) $\lambda(a, b]) = b - a$.
(ii) $\mathcal{B}(\mathbb{R}) \subset \mathcal{L} \subsetneq \mathcal{P}(\mathbb{R})$.
(iii) λ is translation invariant: $\lambda(A + t) = \lambda(A)$ for every $t \in \mathbb{R}$ and $A \in \mathcal{L}$.
(iv) $(\mathbb{R}, \mathcal{L}, \lambda)$ is complete: if $\lambda(N) = 0$ and $A \subseteq N$, then $A \in \mathcal{L}$ (and by monotonicity $\lambda(A) = 0$).

Proof.
(i) The inequality $\lambda^*((a, b]) \leq b - a$ is obtained by taking $(a, b]$ itself as the cover. The other inequality is proved now. Let $\{(a_k, b_k]\}_k$ be a countable cover of $(a, b]$. Given $\epsilon > 0$, enlarge each interval to an *open* one $(a_k - \epsilon_k, b_k + \epsilon_k)$, choosing $\epsilon_k > 0$ so that $\sum_k \epsilon_k < \epsilon$. Thus, $\{(a_k - \epsilon_k, b_k + \epsilon_k)\}$ is an open cover of the compact set $[a, b]$.

By compactness, there is a finite subcover $\{(a_{k_1} - \epsilon_{k_1}, b_{k_1} + \epsilon_{k_1}), \ldots, (a_{k_N} - \epsilon_{k_N}, b_{k_N} + \epsilon_{k_N})\}$ of $[a, b]$. The sum of the lengths of the intervals of the subcover is at least $b - a$:

$$b - a \leq \sum_{j=1}^{N}(b_{k_j} + \epsilon_{k_j} - (a_{k_j} - \epsilon_{k_j})) = \sum_{j=1}^{N}(b_{k_j} - a_{k_j}) + 2\sum_{j=1}^{N}\epsilon_{k_j} \leq \sum_k (b_k - a_k) + 2\epsilon.$$

Letting $\epsilon \to 0$ and then taking the infimum over covers, $b - a \leq \lambda^*((a, b])$.

(ii) It suffices to prove that each half-line (c, ∞) is λ^*-measurable, because the half-lines generate $\mathcal{B}(\mathbb{R})$. Fix $c \in \mathbb{R}$ and $E \subseteq \mathbb{R}$. Given a cover $E \subseteq \bigcup_k (a_k, b_k]$, split each interval into its piece to the right of c and its piece to the left:

$$(a_k, b_k] = ((a_k, b_k] \cap (c, \infty)) \sqcup ((a_k, b_k] \cap (-\infty, c]).$$

Each piece is a half-open interval (possibly empty) and their lengths add up to exactly $b_k - a_k$. Applying λ^* to the two collections of pieces, which cover $E \cap (c, \infty)$ and $E \cap (-\infty, c]$ respectively,

$$\lambda^*(E \cap (c, \infty)) + \lambda^*(E \cap (-\infty, c]) \leq \sum_k (b_k - a_k).$$

Taking the infimum over covers on the right,

$$\lambda^*(E \cap (c, \infty)) + \lambda^*(E \cap (c, \infty)^c) \leq \lambda^*(E).$$

The opposite inequality is subadditivity. Hence (c, ∞) satisfies Carathéodory and belongs to $\mathcal{L}$.

(iii) If $\{(a_k, b_k]\}$ covers E, then $\{(a_k + t, b_k + t]\}$ covers $E + t$ with the same sum of lengths. This gives $\lambda^*(E + t) \leq \lambda^*(E)$, and by symmetry (applying the same to $-t$) equality. Invariance carries

over to λ^*-measurability: if $A \in \mathcal{L}$, Carathéodory's condition for $A+t$ follows by applying that of A to the set $E-t$.

(iv) If $\lambda(N)=0$ and $A \subseteq N$, by monotonicity $\lambda^*(A)=0$. For any E, $\lambda^*(E\cap A) \le \lambda^*(A)=0$, so that Carathéodory's condition for A reduces to $\lambda^*(E)=\lambda^*(E\cap A^c)$, which holds by subadditivity and monotonicity. Hence $A \in \mathcal{L}$.

(v) For $\mathcal{L} \subsetneq \mathcal{P}(\mathbb{R})$, the Vitali sets cannot be Lebesgue-measurable because λ satisfies the three properties in the conclusion of the theorem. □

Remark. There are Lebesgue-null sets with the cardinality of the continuum (Cantor), as well as nowhere dense sets of positive measure (fat Cantor).

Remark. Every countable set has Lebesgue measure zero: for each point, $\lambda(\{x\}) \le \lambda((x-\epsilon, x]) = \epsilon$ for every $\epsilon > 0$, whence $\lambda(\{x\})=0$, and by σ-additivity

$$\lambda(\mathbb{Q}) = \sum_{q\in\mathbb{Q}} \lambda(\{q\}) = 0.$$

Thus, the function $\mathbf{1}_{\mathbb{Q}\cap[0,1]}$ from the motivation, which was not Riemann-integrable, is zero except on a set of measure zero: its Lebesgue integral will be 0.

5.4 Lebesgue-Stieltjes

Definition 5.4 – Lebesgue-Stieltjes outer measure

Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing and right-continuous. For $E \subseteq \mathbb{R}$,

$$\lambda_F^*(E) = \inf\left\{ \sum_{n=1}^{\infty} (F(b_n) - F(a_n)) : E \subseteq \bigcup_n (a_n, b_n] \right\}.$$

Theorem 5.3 – Lebesgue-Stieltjes measure

Let F be as above. Then λ_F^* is an outer measure, and the Carathéodory σ-algebra $\mathcal{L}_F$ contains $\mathcal{B}(\mathbb{R})$. The restriction $\lambda_F = \lambda_F^*|_{\mathcal{L}_F}$ is the *Lebesgue-Stieltjes measure* of F, and it satisfies

$$\lambda_F((a,b]) = F(b) - F(a) \qquad \forall a < b.$$

Proof. The identity $\lambda_F((a,b]) = F(b)-F(a)$ uses, in the compactness step, that F is right-continuous: to show $F(b)-F(a) \le \sum_k (F(b_k)-F(a_k))$ when $\{(a_k,b_k]\}$ covers $(a,b]$, the intervals are widened to $(a_k, b_k+\delta_k]$ with $F(b_k+\delta_k)-F(b_k) < \epsilon/2^k$ and $[a+\delta, b]$ is compactified as for Lebesgue. The λ_F^*-measurability of the half-lines and completeness follow exactly in the same way. □

Remark. When F is the distribution function of a random variable X, λ_F is the distribution of X, denoted P_X, and $P_X(B) = P(X \in B)$ for every Borel set B.

6 Measurable functions

Definition 6.1 – Measurable function

Let $(X, \mathcal{A}, \mu)$ be a measure space. A function $f : X \to \mathbb{R}$ is *measurable* if for every $a \in \mathbb{R}$,

$$f^{-1}((a, \infty)) = \{x \in X : f(x) > a\} \in \mathcal{A}.$$

Remark. In probability, a measurable function is called a *random variable*.

Remark. The definition points directly at the Lebesgue integral: to approximate $\int f$ by partitioning the *range* into values $a_0 < a_1 < \cdots < a_n$, one must measure the sets

$$A_k = \{a_{k-1} < f \le a_k\} = \{f > a_{k-1}\} \setminus \{f > a_k\}.$$

Requiring $\{f > a\} \in \mathcal{A}$ for every a is exactly what guarantees that those sets have a measure.

Proposition 6.1 – Equivalent characterizations of measurability

Let $(X, \mathcal{A})$ be a measurable space and $f : X \to \mathbb{R}$. The following conditions are equivalent:

(i) $\{f > a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.
(ii) $\{f \ge a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.
(iii) $\{f < a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.
(iv) $\{f \le a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.

Proof.
(i) $\Leftrightarrow$ (iv), (ii) $\Leftrightarrow$ (iii) and (i) $\Leftrightarrow$ (ii) are proved.

(i) $\Leftrightarrow$ (iv). $\{f \le a\} = \{f > a\}^c$, and $\mathcal{A}$ is closed under complements.

(ii) $\Leftrightarrow$ (iii). $\{f < a\} = \{f \ge a\}^c$.

(i) $\Rightarrow$ (ii). $\{f \ge a\} = \bigcap_{n=1}^{\infty} \{f > a - 1/n\}$, a countable intersection of elements of $\mathcal{A}$.

(ii) $\Rightarrow$ (i). $\{f > a\} = \bigcup_{n=1}^{\infty} \{f \ge a + 1/n\}$, a countable union of elements of $\mathcal{A}$.

With this, (i)–(iv) are equivalent. □

6.1 Examples and operations

Example 6.1 – Constant functions

If $f \equiv c$, then $\{f > a\}$ equals X if $c > a$ and $\emptyset$ if $c \le a$. Since $X, \emptyset \in \mathcal{A}$, f is measurable.

Example 6.2 – Characteristic functions

Let $A \in \mathcal{A}$. The characteristic function $\mathbf{1}_A$ is measurable:

$$\{\mathbf{1}_A > a\} = \begin{cases} X, & a < 0, \\ A, & 0 \le a < 1, \\ \emptyset, & a \ge 1, \end{cases}$$

and the three sets are in $\mathcal{A}$.

Theorem 6.1 – Continuous functions are Borel-measurable

Every continuous function $f : \mathbb{R} \to \mathbb{R}$ is Borel-measurable.

Proof. It must be seen that $f^{-1}((a, \infty)) \in \mathcal{B}(\mathbb{R})$ for every $a \in \mathbb{R}$. Since (a, ∞) is open and f continuous, $f^{-1}((a, \infty))$ is open in $\mathbb{R}$, and every open set belongs to $\mathcal{B}(\mathbb{R})$.

More generally, the preimage under a continuous function of any open set is open, which gives $f^{-1}(\mathcal{B}(\mathbb{R})) \subseteq \mathcal{B}(\mathbb{R})$: the set $\{B \subseteq \mathbb{R} : f^{-1}(B) \in \mathcal{B}(\mathbb{R})\}$ is a σ-algebra containing the open sets, so that it contains $\mathcal{B}(\mathbb{R})$. □

Theorem 6.2 – Monotone functions are Borel-measurable

Every monotone function $f : \mathbb{R} \to \mathbb{R}$ is Borel-measurable.

Proof. Suppose f nondecreasing (the nonincreasing case follows by applying what follows to $-f$). With $a \in \mathbb{R}$ fixed, consider the set $E_a = f^{-1}((a, \infty)) = \{x \in \mathbb{R} : f(x) > a\}$.

If $x \in E_a$ and $y > x$, monotonicity gives $f(y) \geq f(x) > a$, so that $y \in E_a$. Therefore E_a is a half-line to the right: denoting $c = \inf E_a$ (with the convention $\inf \emptyset = +\infty$, and inf of a set not bounded below $= -\infty$), then

$$E_a = (c, \infty) \quad \text{or} \quad E_a = [c, \infty),$$

according to whether the infimum is attained or not. In either case, $E_a \in \mathcal{B}(\mathbb{R})$ (open and closed half-lines are Borel sets). □

Example 6.3 – Simple functions

A function $\varphi : X \to \mathbb{R}$ is *simple* if it is a finite linear combination of characteristic functions:

$$\varphi = \sum_{i=1}^{n} c_i \, \mathbf{1}_{A_i}, \qquad c_i \in \mathbb{R}, \; A_i \in \mathcal{A}.$$

Equivalently, φ takes only finitely many values and the preimage of each one is in $\mathcal{A}$. Simple functions are measurable because each $\mathbf{1}_{A_i}$ is and the operations of sum and product by a scalar preserve measurability (Theorem 6.3).

Example 6.4 – Canonical representation of a simple function

Writing a simple function as a combination of characteristic functions is not unique. With intervals of $\mathbb{R}$, consider

$$\varphi = 2 \cdot \mathbf{1}_{[0,2]} + 3 \cdot \mathbf{1}_{[1,5]}.$$

The two intervals overlap on $[1, 2]$, where φ equals $2 + 3 = 5$. Separating according to the values that φ takes,

$$\varphi = 2 \cdot \mathbf{1}_{[0,1)} + 5 \cdot \mathbf{1}_{[1,2]} + 3 \cdot \mathbf{1}_{(2,5]},$$

which is the *canonical representation*: the coefficients are the distinct nonzero values of φ, and the sets $\{\varphi = 2\} = [0, 1)$, $\{\varphi = 5\} = [1, 2]$ and $\{\varphi = 3\} = (2, 5]$ are disjoint. Every simple function admits a unique representation of this form, which is the one used by the definition of the integral.

Theorem 6.3 – Operations with measurable functions

Let $f, g : X \to \mathbb{R}$ be measurable and $c \in \mathbb{R}$. Then cf, $f+g$, $f \cdot g$, $\max(f,g)$, $\min(f,g)$, $|f|$, $f^+ := \max(f,0)$ and $f^- := \max(-f,0)$ are measurable.

Proof.
By Proposition 6.1, to see that a function is measurable it suffices to verify any of the conditions $\{f > a\}, \{f \geq a\}, \{f < a\}, \{f \leq a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$; in each case the most convenient one is chosen.

Product by a scalar.

- If $c = 0$, $cf \equiv 0$, which is constant and therefore measurable.
- If $c > 0$, $cf(x) > a \Leftrightarrow f(x) > a/c$, so that $\{cf > a\} = \{f > a/c\} \in \mathcal{A}$.
- If $c < 0$, $cf(x) > a \Leftrightarrow f(x) < a/c$, so that $\{cf > a\} = \{f < a/c\} \in \mathcal{A}$.

Maximum and minimum. Since

$$\max(f,g)(x) > a \;\Leftrightarrow\; f(x) > a \;\text{ or }\; g(x) > a,$$

one has $\{\max(f,g) > a\} = \{f > a\} \cup \{g > a\} \in \mathcal{A}$, being a finite union of measurable sets. Analogously,

$$\min(f,g)(x) \geq a \;\Leftrightarrow\; f(x) \geq a \;\text{ and }\; g(x) \geq a,$$

so that $\{\min(f,g) \geq a\} = \{f \geq a\} \cap \{g \geq a\} \in \mathcal{A}$.

Absolute value. For $a < 0$, $\{|f| > a\} = X \in \mathcal{A}$. For $a \geq 0$,

$$|f(x)| > a \;\Leftrightarrow\; f(x) > a \;\text{ or }\; f(x) < -a,$$

so that $\{|f| > a\} = \{f > a\} \cup \{f < -a\} \in \mathcal{A}$.

Positive and negative parts f^+, f^-. The constant function 0 is measurable, and by the case of the maximum applied to f and 0 (or to $-f$ and 0, using the case of product by a scalar with $c = -1$),

$$f^+ = \max(f,0), \qquad f^- = \max(-f,0),$$

are measurable. Moreover the identities $f = f^+ - f^-$ and $|f| = f^+ + f^-$ hold.

Sum. It is shown that for every $a \in \mathbb{R}$,

$$\{f+g > a\} = \bigcup_{r \in \mathbb{Q}} \left(\{f > r\} \cap \{g > a - r\}\right).$$

- ($\subseteq$) Let x be such that $f(x) + g(x) > a$, that is $f(x) > a - g(x)$. By density of $\mathbb{Q}$ in $\mathbb{R}$, there is $r \in \mathbb{Q}$ with
$$a - g(x) < r < f(x).$$
From $r < f(x)$ one gets $x \in \{f > r\}$, and from $a - g(x) < r$ one gets $g(x) > a - r$, that is $x \in \{g > a - r\}$.
- ($\supseteq$) Let $r \in \mathbb{Q}$ with $f(x) > r$ and $g(x) > a - r$. Adding, $f(x) + g(x) > r + (a - r) = a$, that is $x \in \{f + g > a\}$.

The right-hand side is a *countable* union of finite intersections of sets in $\mathcal{A}$, and therefore belongs to $\mathcal{A}$.

Square. For $a < 0$, $\{f^2 > a\} = X \in \mathcal{A}$. For $a \geq 0$,

$$f(x)^2 > a \;\Leftrightarrow\; |f(x)| > \sqrt{a} \;\Leftrightarrow\; f(x) > \sqrt{a} \;\text{ or }\; f(x) < -\sqrt{a},$$

and therefore $\{f^2 > a\} = \{f > \sqrt{a}\} \cup \{f < -\sqrt{a}\} \in \mathcal{A}$.

Product. The polarization identity is used:

$$fg = \tfrac{1}{4}\left[(f+g)^2 - (f-g)^2\right].$$

By the previous steps, $f+g$ and $f-g = f+(-1)g$ are measurable. Their squares $(f+g)^2$ and $(f-g)^2$ are measurable by the previous case. Applying sum and product by a scalar once more,

$$fg = \tfrac{1}{4}(f+g)^2 + \left(-\tfrac{1}{4}\right)(f-g)^2 \in \mathcal{A},$$

that is, fg is measurable. □

Remark. The positive and negative parts satisfy, in addition to $f = f^+ - f^-$ and $|f| = f^+ + f^-$, the inverse formulas

$$f^+ = \tfrac{1}{2}(|f| + f), \qquad f^- = \tfrac{1}{2}(|f| - f),$$

with $f^+, f^- \geq 0$ of disjoint supports. The decomposition $f = f^+ - f^-$ is what reduces the integral of a general function to that of two nonnegative ones.

6.2 Sup, inf, lim sup, lim inf

Theorem 6.4 – Sup and inf of measurable functions

If (f_n) is a sequence of measurable functions, then $\sup_n f_n$, $\inf_n f_n$, $\overline{\lim}_n f_n$ and $\underline{\lim}_n f_n$ are measurable.

Proof.
Case of the supremum. For every $a \in \mathbb{R}$,

$$\sup_n f_n(x) > a \;\Leftrightarrow\; \exists\, n \in \mathbb{N} : f_n(x) > a,$$

whence

$$\{\sup_n f_n > a\} = \bigcup_{n\in\mathbb{N}} \{f_n > a\}.$$

Each $\{f_n > a\} \in \mathcal{A}$ by measurability of f_n, and the union is countable, so that $\{\sup_n f_n > a\} \in \mathcal{A}$ for every a.

Case of the infimum. Analogously,

$$\inf_n f_n(x) < a \;\Leftrightarrow\; \exists\, n \in \mathbb{N} : f_n(x) < a,$$

so that $\{\inf_n f_n < a\} = \bigcup_n \{f_n < a\} \in \mathcal{A}$.

Limit superior. By definition,

$$\overline{\lim_n} f_n = \inf_{m\in\mathbb{N}} \sup_{n\geq m} f_n.$$

For each m, the sequence $(f_n)_{n\geq m}$ is measurable, so that $g_m := \sup_{n\geq m} f_n$ is measurable by the case of the sup. The sequence $(g_m)_m$ is then measurable, and $\overline{\lim}_n f_n = \inf_m g_m$ is measurable by the case of the inf.

Limit inferior. Analogous, exchanging sup with inf: $\underline{\lim}_n f_n = \sup_m \inf_{n\geq m} f_n$, where $\inf_{n\geq m} f_n$ is measurable by the case of the inf and then the sup over m by the case of the sup. □

Remark. The four operations $\sup, \inf, \overline{\lim}, \underline{\lim}$ may yield functions with values in $[-\infty, +\infty]$. The definition of measurability extends without problems: $f : X \to [-\infty, +\infty]$ is measurable if $\{f > a\} \in \mathcal{A}$ for every $a \in \mathbb{R}$.

Corollary 6.1 – Pointwise limit of measurable functions is measurable

Let (f_n) be a sequence of measurable functions.
(i) If $f_n \to f$ pointwise on all of X, f is measurable.
(ii) If the series $g := \sum_{k=1}^{\infty} f_k$ converges pointwise, g is measurable.

Proof.
(i) If $f_n \to f$ pointwise, $f = \overline{\lim}_n f_n = \underline{\lim}_n f_n$, and by the previous theorem both are measurable. Hence f is measurable.

(ii) The partial sums $g_n = \sum_{k=1}^{n} f_k$ are finite sums of measurable functions, and by Theorem 6.3 they are measurable. Since $g = \lim_n g_n$ pointwise (by hypothesis), by (i) g is measurable. □

6.3 Approximation by simple functions

Theorem 6.5 – Approximation by simple functions

Let $f : X \to \mathbb{R}$ be measurable and nonnegative. There is a sequence (φ_n) of nonnegative simple functions such that $\varphi_n \nearrow f$ pointwise. If moreover f is bounded, the convergence is uniform.

Proof.
The idea is to discretize the *values* of f. For $n \in \mathbb{N}$ and $0 \leq k \leq n2^n - 1$, write $I_{n,k} = [k/2^n, (k+1)/2^n)$ and define

$$\varphi_n(x) = \begin{cases} \dfrac{k}{2^n}, & \text{if } f(x) \in I_{n,k}, \\ n, & \text{if } f(x) \geq n. \end{cases}$$

The intervals $I_{n,0}, \ldots, I_{n,n2^n-1}$ together with $[n, \infty)$ form a partition of $[0, \infty)$, so that φ_n is well defined and takes at most $n2^n+1$ values. Its level sets are the preimages $\{\varphi_n = k/2^n\} = f^{-1}(I_{n,k})$ and $\{\varphi_n = n\} = f^{-1}([n, \infty))$, measurable because f is; hence φ_n is simple. On the other hand, $\varphi_n(x) \leq f(x)$ in each case ($k/2^n \leq f(x)$ when $f(x) \in I_{n,k}$, and $n \leq f(x)$ when $f(x) \geq n$), so that $0 \leq \varphi_n \leq f$.

It is shown that $\varphi_n \leq \varphi_{n+1}$ pointwise. Since $I_{n,k} = I_{n+1,2k} \sqcup I_{n+1,2k+1}$, if $f(x) \in I_{n,k}$ with $k \leq n2^n - 1$ then $f(x)$ belongs to one of those two intervals, on which $\varphi_{n+1}(x)$ equals $2k/2^{n+1} = k/2^n$ or $(2k+1)/2^{n+1} > k/2^n$; in both cases $\varphi_{n+1}(x) \geq k/2^n = \varphi_n(x)$. If instead $f(x) \geq n$, then $\varphi_n(x) = n$ and $\varphi_{n+1}(x) \geq n$: it equals $n+1$ if $f(x) \geq n+1$, and if $n \leq f(x) < n+1$

there is k with $f(x) \in I_{n+1,k}$ and $k/2^{n+1} \geq n$, whence $\varphi_{n+1}(x) = k/2^{n+1} \geq n$. In all cases $\varphi_{n+1}(x) \geq \varphi_n(x)$.

Now let x be such that $f(x) < \infty$ and take $n > f(x)$. Then $f(x) < n$, so that $f(x) \in I_{n,k}$ for some k and, by definition of $I_{n,k}$,

$$0 \leq f(x) - \varphi_n(x) = f(x) - \frac{k}{2^n} < \frac{1}{2^n}.$$

Since $2^{-n} \to 0$, one obtains $\varphi_n(x) \to f(x)$.

Finally, if $f \leq M$, for every $n > M$ one has $f(x) < n$ whatever x is, so that the bound $0 \leq f(x) - \varphi_n(x) < 2^{-n}$ holds uniformly in x. Since $2^{-n} \to 0$, the convergence is uniform. □

7 Convergence of measurable functions

Definition 7.1 – Convergence almost everywhere

$f_n \to f$ *almost everywhere* (a.e.) if there is $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $f_n(x) \to f(x)$ for every $x \notin N$.

Definition 7.2 – Convergence in measure

$f_n \to f$ *in measure* if for every $\epsilon > 0$,

$$\mu(\{x : |f_n(x) - f(x)| \geq \epsilon\}) \xrightarrow{n\to\infty} 0.$$

Definition 7.3 – Almost uniform convergence

$f_n \to f$ *almost uniformly* if for every $\delta > 0$ there is $N \in \mathcal{A}$ with $\mu(N) < \delta$ such that $f_n \to f$ uniformly on N^c.

Corollary 7.1 – A.e. limit of measurable functions

If $\mathcal{A}$ is complete with respect to μ, the f_n are measurable and $f_n \to f$ a.e., then f is measurable.

Proof. There is $N \in \mathcal{A}$ with $\mu(N) = 0$ such that $f_n(x) \to f(x)$ for every $x \notin N$. Let $\tilde{f} = \overline{\lim}_n f_n$, measurable by Theorem 6.4; on N^c the sequence converges, so that $f = \tilde{f}$ on N^c. With $a \in \mathbb{R}$ fixed, decompose

$$\{f > a\} = \big(\{f > a\} \cap N^c\big) \cup \big(\{f > a\} \cap N\big).$$

The first set coincides with $\{\tilde{f} > a\} \cap N^c$, an intersection of two measurable sets. The second satisfies $\{f > a\} \cap N \subseteq N$, and since $\mathcal{A}$ is complete, every subset of a μ-null set is measurable. Therefore $\{f > a\}$ is a union of two measurable sets, and f is measurable. □

Remark. Without completeness of $\mathcal{A}$, a.e. convergence does not guarantee measurability: if $\mu(N) = 0$ but N contains non-measurable subsets, redefining f arbitrarily on N can break measurability.

Remark. In general, in any measure space the implications

$$\text{uniform} \implies \text{pointwise} \implies \text{a.e.}, \qquad \text{almost uniform} \implies \text{in measure},$$

hold, and almost uniform convergence also implies a.e. convergence. But *no* other implication holds automatically. Typical examples:

- $f_n = \mathbf{1}_{[n,n+1]}$ on $\mathbb{R}$ converges pointwise to 0, but *not* in measure.
- The functions $\mathbf{1}_{[0,1]}, \mathbf{1}_{[0,1/2]}, \mathbf{1}_{[1/2,1]}, \mathbf{1}_{[0,1/4]}, \ldots$ converge to 0 in measure but *not* pointwise at any point.
- $f_n(x) = x^n$ on $([0,1], \lambda)$ converges pointwise to $\mathbf{1}_{\{1\}}$ but *not* uniformly: given $0 < \epsilon < 1$ and $x < 1$, the condition $x^n < \epsilon$ requires $n > \ln \epsilon / \ln x$, a bound that blows up as $x \to 1^-$, so that no n_0 works for all x at once (observe moreover that the pointwise limit is not continuous even though each f_n is). It does converge *almost uniformly* to $\mathbf{1}_{\{1\}}$: given

$\delta > 0$ it suffices to exclude $N = (1-\delta, 1)$, and on $[0, 1-\delta]$ the convergence to 0 is uniform because $x^n \le (1-\delta)^n \to 0$.

7.1 Egoroff's theorem

Theorem 7.1 – Egoroff

Let $(X, \mathcal{A}, \mu)$ be a finite measure space ($\mu(X) < \infty$) and $f_n \to f$ a.e. Then:
(i) $f_n \to f$ almost uniformly (Definition 7.3).
(ii) As a consequence, $f_n \to f$ in measure.

Proof. Fix $\delta > 0$. For each $p, m \in \mathbb{N}$, let

$$A_m^{(p)} = \{x \in X : |f_m(x) - f(x)| \ge 1/p\}.$$

It is first shown that, for each p,

$$\mu\left(\bigcup_{k \ge m} A_k^{(p)}\right) \xrightarrow{m \to \infty} 0.$$

Indeed, if $x \in \overline{\lim}_m A_m^{(p)} = \bigcap_m \bigcup_{k \ge m} A_k^{(p)}$, then for each m there is $k \ge m$ with $|f_k(x) - f(x)| \ge 1/p$, whence $f_k(x) \not\to f(x)$. Therefore $\overline{\lim}_m A_m^{(p)} \subseteq \{x \in X : f_n(x) \not\to f(x)\}$, which is μ-null by hypothesis. Since moreover $\left(\bigcup_{k \ge m} A_k^{(p)}\right)_m$ decreases to $\overline{\lim}_m A_m^{(p)}$ and its first term has measure $\le \mu(X) < \infty$, continuity of the measure gives the announced limit.

(i) is proved. By what was just seen, for each $p \in \mathbb{N}$ one can choose m_p such that $\mu\big(\bigcup_{k \ge m_p} A_k^{(p)}\big) < \delta/2^p$, and define

$$N = \bigcup_{p=1}^{\infty} \bigcup_{k \ge m_p} A_k^{(p)}, \qquad \text{with } \mu(N) \le \sum_{p=1}^{\infty} \frac{\delta}{2^p} = \delta.$$

If $x \in N^c$, then $|f_k(x) - f(x)| < 1/p$ for every $p \in \mathbb{N}$ and every $k \ge m_p$. Thus, given $\epsilon > 0$ and choosing p with $1/p < \epsilon$, one gets $|f_k(x) - f(x)| < \epsilon$ for every $k \ge m_p$ and every $x \in N^c$. Since m_p does not depend on x, the convergence is uniform on N^c.

Finally, (ii) follows from (i): given $\epsilon > 0$ and $\eta > 0$, by almost uniform convergence there are N with $\mu(N) < \eta$ and $m_0 \in \mathbb{N}$ such that $|f_m - f| < \epsilon$ on N^c for every $m \ge m_0$. Then $\{|f_m - f| \ge \epsilon\} \subseteq N$ for $m \ge m_0$, whence $\mu(\{|f_m - f| \ge \epsilon\}) \le \mu(N) < \eta$. Since η is arbitrary, $\mu(\{|f_m - f| \ge \epsilon\}) \to 0$; that is, $f_m \to f$ in measure. □

Remark. The m_p that defines N depends on p but *not* on x.

8 Integration

The integral is constructed in three steps: simple, nonnegative, general.

8.1 Step 1: nonnegative simple functions

Definition 8.1 – Integral of a nonnegative simple function

Let $\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}$ be simple with $a_i \geq 0$ and A_i disjoint. Define

$$\int_X \varphi \, d\mu = \sum_{i=1}^{n} a_i \, \mu(A_i),$$

with the convention $0 \cdot \infty = 0$.

Remark. The definition does not depend on the representation (it suffices to verify it for refinements).

Proposition 8.1 – Linearity and monotonicity on simple functions

If φ, ψ are nonnegative simple functions and $a, b \geq 0$:
(i) $\int (a\varphi + b\psi) d\mu = a \int \varphi \, d\mu + b \int \psi \, d\mu$.
(ii) If $\varphi \leq \psi$, $\int \varphi \, d\mu \leq \int \psi \, d\mu$.
Moreover, for $A \in \mathcal{A}$, $\lambda(A) := \int_A \varphi \, d\mu := \int_X \varphi \mathbf{1}_A \, d\mu$ is a measure on $(X, \mathcal{A})$.

Proof.
(i) Refining both representations to a common partition, one may suppose that $\varphi = \sum_i \alpha_i \mathbf{1}_{C_i}$ and $\psi = \sum_i \beta_i \mathbf{1}_{C_i}$ use the same disjoint partition $\{C_i\}$. Then $a\varphi + b\psi = \sum_i (a\alpha_i + b\beta_i)\mathbf{1}_{C_i}$, and by definition,

$$\int (a\varphi + b\psi) d\mu = \sum_i (a\alpha_i + b\beta_i)\mu(C_i) = a \sum_i \alpha_i \mu(C_i) + b \sum_i \beta_i \mu(C_i) = a \int \varphi \, d\mu + b \int \psi \, d\mu.$$

(ii) With the same common partition, $\varphi \leq \psi$ implies $\alpha_i \leq \beta_i$ on each nonempty C_i. Summing with $\mu(C_i) \geq 0$ preserves the inequality: $\int \varphi \, d\mu \leq \int \psi \, d\mu$.

λ is a measure. $\lambda(\emptyset) = 0$ is direct. For σ-additivity, let (A_n) be disjoint and $A = \bigsqcup_n A_n$. With $\varphi = \sum_i a_i \mathbf{1}_{C_i}$ in canonical form,

$$\lambda(A) = \sum_i a_i \mu(C_i \cap A) = \sum_i a_i \sum_n \mu(C_i \cap A_n) = \sum_n \sum_i a_i \mu(C_i \cap A_n) = \sum_n \lambda(A_n),$$

where the interchange of sums is legitimate since all terms are nonnegative (Tonelli's theorem for series). □

8.2 Step 2: nonnegative measurable functions

Definition 8.2 – Integral of a nonnegative function

Let $f : X \to [0, \infty]$ be measurable. Define

$$\int_X f \, d\mu = \sup \left\{ \int_X \varphi \, d\mu : \varphi \text{ nonnegative simple}, \, 0 \le \varphi \le f \right\}.$$

Proposition 8.2 – Basic properties for nonnegative functions

For measurable $f, g \ge 0$:

(i) If $f \le g$, $\int f \, d\mu \le \int g \, d\mu$.
(ii) If $A \subseteq B$ are measurable, $\int_A f \, d\mu \le \int_B f \, d\mu$.
(iii) If $A \in \mathcal{A}$, $\nu(A) := \int_A f \, d\mu$ defines a measure on $(X, \mathcal{A})$.

Proof.
(i) Every nonnegative simple φ with $0 \le \varphi \le f$ also satisfies $0 \le \varphi \le g$, so that the supremum defining $\int f$ is taken over a subfamily of the one that defines $\int g$. Therefore $\int f \le \int g$.

(ii) $f\mathbf{1}_A \le f\mathbf{1}_B$ because $\mathbf{1}_A \le \mathbf{1}_B$. By (i), $\int_A f = \int f\mathbf{1}_A \le \int f\mathbf{1}_B = \int_B f$.

(iii) One has $\nu(\emptyset) = 0$. For σ-additivity, let (A_n) be disjoint and $A = \bigsqcup_n A_n$.

Lower bound $\nu(A) \ge \sum_n \nu(A_n)$. By (ii), for every finite N, $\sum_{n=1}^N \nu(A_n) = \nu\left(\bigsqcup_{n=1}^N A_n\right) \le \nu(A)$, where the first equality follows from finite additivity: for a simple $\varphi \le f$, $\int \varphi \mathbf{1}_{\bigsqcup_{n=1}^N A_n} = \sum_{n=1}^N \int \varphi \mathbf{1}_{A_n}$ by σ-additivity of μ applied to the measure defined by φ (Proposition 8.1). Letting $N \to \infty$, $\sum_{n=1}^\infty \nu(A_n) \le \nu(A)$.

Upper bound $\nu(A) \le \sum_n \nu(A_n)$. Let φ be nonnegative simple with $0 \le \varphi \le f\mathbf{1}_A$. Since φ vanishes outside A, $\varphi = \sum_n \varphi \mathbf{1}_{A_n}$ (a pointwise sum where at most one term is nonzero by disjointness). By σ-additivity of the measure $E \mapsto \int_E \varphi \, d\mu$ (again Proposition 8.1),

$$\int_A \varphi \, d\mu = \sum_n \int_{A_n} \varphi \, d\mu \le \sum_n \nu(A_n),$$

because each $\int_{A_n} \varphi = \int \varphi \mathbf{1}_{A_n} \le \int f \mathbf{1}_{A_n} = \nu(A_n)$. Taking the supremum over φ, $\nu(A) \le \sum_n \nu(A_n)$.

Combining both inequalities, $\nu(A) = \sum_n \nu(A_n)$, that is, ν is σ-additive. □

8.3 Step 3: general functions

Definition 8.3 – Integrable function

Let $f : X \to \mathbb{R}$ be measurable. Decompose $f = f^+ - f^-$, where $f^\pm \ge 0$. f is said to be *integrable* if $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$, and define

$$\int_X f \, d\mu = \int_X f^+ d\mu - \int_X f^- d\mu.$$

Remark. If only one of the two parts is finite, the integral is defined with value in $\mathbb{R} \cup \{\pm\infty\}$. If both are infinite, the integral is not defined.

Proposition 8.3 – Characterization by $|f|$

f is integrable if and only if $|f|$ is integrable, and in that case

$$\left| \int_X f \, d\mu \right| \leq \int_X |f| \, d\mu \qquad \text{(triangle inequality)}.$$

If $|f| \leq g$ with g integrable, then f is integrable.

Proof. $|f| = f^+ + f^-$, so that $|f|$ is integrable iff f^+ and f^- are. For the triangle inequality, $|\int f| = |\int f^+ - \int f^-| \leq \int f^+ + \int f^- = \int |f|$. The last claim follows by monotonicity. □

Proposition 8.4 – Linearity and monotonicity

If f, g are integrable and $a, b \in \mathbb{R}$:

(i) $af + bg$ is integrable and $\int (af + bg) d\mu = a \int f \, d\mu + b \int g \, d\mu$.

(ii) If $f \leq g$, $\int f \, d\mu \leq \int g \, d\mu$.

(iii) $f = 0$ a.e. $\implies \int f \, d\mu = 0$ (and the converse for $f \geq 0$: $\int f \, d\mu = 0 \implies f = 0$ a.e.).

Proof of (iii). The converse is proved: if $f \geq 0$ and $\int f \, d\mu = 0$, then $f = 0$ a.e. For each m, the set $A_m = \{f \geq 1/m\}$ satisfies $f \geq \frac{1}{m} \mathbf{1}_{A_m}$, whence, integrating,

$$\frac{1}{m} \mu(A_m) \leq \int_X f \, d\mu = 0.$$

Hence $\mu(A_m) = 0$ for every m, and since $\{f > 0\} = \bigcup_m A_m$, it follows that $\mu(\{f > 0\}) = 0$. □

Remark. For signed functions, $\int_X f \, d\mu = 0$ does not imply $f = 0$ a.e. (take an integrable odd f on $(\mathbb{R}, \lambda)$). The correct version requires all the restrictions: if

$$\int_A f \, d\mu = 0 \qquad \text{for every } A \in \mathcal{A},$$

then $f = 0$ a.e. Indeed, with $A = \{f \geq 0\}$ one gets $f \cdot \mathbf{1}_A = f^+$, whence $\int_X f^+ \, d\mu = 0$ and $f^+ = 0$ a.e. by the previous result; with $A = \{f \leq 0\}$ one gets analogously $f^- = 0$ a.e. This remark is what gives uniqueness in the Radon-Nikodym theorem and in conditional expectation.

8.4 Classical inequalities

Theorem 8.1 – Markov's inequality

Let $f \geq 0$ be measurable. For every $a > 0$,

$$\mu(\{f \geq a\}) \leq \frac{1}{a} \int_X f \, d\mu.$$

Proof. $f \geq a \cdot \mathbf{1}_{\{f \geq a\}}$, and integrating $\int f \, d\mu \geq a\mu(\{f \geq a\})$. □

Corollary 8.1 – Chebyshev's inequality

If $X \in L^p$ with $1 \le p < \infty$, for every $a > 0$,

$$\mu(\{|X| \ge a\}) \le \frac{1}{a^p} \int |X|^p \, d\mu = \frac{\|X\|_p^p}{a^p}.$$

Proof. Apply Markov to $f = |X|^p$ with threshold a^p. □

Remark. In probability: $P(|X - E(X)| \ge a) \le \mathrm{Var}(X)/a^2$ (Chebyshev with $p = 2$ applied to $X - E(X)$). It is the basis of the weak law of large numbers.

Theorem 8.2 – Jensen's inequality (integral)

Let μ be a probability ($\mu(X) = 1$), $f \in L^1(\mu)$, and $\varphi : \mathbb{R} \to \mathbb{R}$ convex with $\varphi \circ f$ integrable. Then

$$\varphi\left(\int_X f \, d\mu\right) \le \int_X \varphi \circ f \, d\mu.$$

Proof. Let $c = \int_X f \, d\mu$. Since φ is convex, at each point it admits a supporting line: there is $m \in \mathbb{R}$ such that

$$\varphi(y) \ge \varphi(c) + m(y - c) \qquad \forall y \in \mathbb{R}.$$

(The m is any value between the one-sided derivatives $\varphi'_-(c)$ and $\varphi'_+(c)$, which exist by convexity.) Replacing $y = f(x)$ and integrating with respect to μ,

$$\int_X \varphi \circ f \, d\mu \ge \int_X \varphi(c) \, d\mu + m \int_X (f - c) \, d\mu = \varphi(c)\mu(X) + m \left(\int f \, d\mu - c\mu(X) \right).$$

Since $\mu(X) = 1$ and $\int f \, d\mu = c$, the second term is zero and one gets $\int \varphi \circ f \, d\mu \ge \varphi(c) = \varphi(\int f \, d\mu)$. □

Remark. Useful particular cases: $\varphi(t) = t^p$ for $p \ge 1$ gives $|\int f|^p \le \int |f|^p$ (if μ is a probability); $\varphi(t) = e^t$ gives $\exp(\int f) \le \int e^f$; $\varphi(t) = -\ln t$ (with $t > 0$) gives $\ln \int f \ge \int \ln f$.

9 Convergence theorems

When does $\lim_n \int f_n = \int \lim_n f_n$ hold? Not always: let $f_n = n \cdot \mathbf{1}_{(0,1/n)}$ on $(\mathbb{R}, \lambda)$. Then $f_n \to 0$ pointwise (with $x > 0$ fixed, for $n > 1/x$ one has $f_n(x) = 0$; at $x \le 0$ all of them vanish), but

$$\int_{\mathbb{R}} f_n \, d\lambda = n \cdot \frac{1}{n} = 1 \qquad \text{for every } n,$$

so that $\lim_n \int f_n = 1 \neq 0 = \int \lim_n f_n$: the mass escapes through an increasingly tall and narrow spike. The same phenomenon in the version for series: with $g_1 = f_1$ and $g_n = f_n - f_{n-1}$ one has $\sum_{n=1}^{m} g_n = f_m$, and

$$\sum_{n=1}^{\infty} \int_{\mathbb{R}} g_n \, d\lambda \neq \int_{\mathbb{R}} \sum_{n=1}^{\infty} g_n \, d\lambda.$$

The theorems of this section give conditions (monotonicity, domination) under which the interchange is legitimate.

Theorem 9.1 – Monotone Convergence (Beppo Levi)

Let (f_n) be an increasing sequence of nonnegative measurable functions with $f_n \nearrow f$ pointwise. Then

$$\int_X f \, d\mu = \lim_{n\to\infty} \int_X f_n \, d\mu.$$

Proof. $f = \lim f_n$ is measurable. Since $f_n \le f$ for every n,

$$\lim_n \int f_n \, d\mu \le \int f \, d\mu.$$

For the other inequality, let φ be nonnegative simple with $0 \le \varphi \le f$ and $0 < c < 1$. Define $A_n = \{x : f_n(x) \ge c\varphi(x)\}$. Since $f_n \nearrow f \ge \varphi > c\varphi$ pointwise on $\{\varphi > 0\}$, the A_n increase to X (or, more precisely, their union covers $\{\varphi > 0\}$, and outside it there is nothing to prove). Then

$$\int f_n \, d\mu \ge \int_{A_n} f_n \, d\mu \ge c \int_{A_n} \varphi \, d\mu.$$

Since $\nu(A) := \int_A \varphi \, d\mu$ is a measure, by continuity $\int_{A_n} \varphi \, d\mu \to \int_X \varphi \, d\mu$. Passing to the limit, $\lim_n \int f_n \, d\mu \ge c \int \varphi \, d\mu$. Letting $c \to 1$ and then taking the sup over φ, one gets $\lim_n \int f_n \, d\mu \ge \int f \, d\mu$. □

Theorem 9.2 – Fatou's Lemma

If (f_n) is a sequence of nonnegative measurable functions,

$$\int_X \varliminf_n f_n \, d\mu \le \varliminf_n \int_X f_n \, d\mu.$$

Proof. Define $g_m = \inf_{n \ge m} f_n$. The sequence (g_m) is increasing and $g_m \nearrow \varliminf_n f_n$. By MCT,

$$\int_X \varliminf_n f_n \, d\mu = \lim_m \int_X g_m \, d\mu.$$

Since $g_m \le f_n$ for every $n \ge m$, $\int g_m \le \int f_n$, so that $\int g_m \le \inf_{n \ge m} \int f_n$. Passing to the limit in m, $\lim_m \int g_m \le \varliminf_n \int f_n$. □

Remark. In general the inequality is strict. Example: $f_n = \mathbf{1}_{[n,n+1]}$ on $\mathbb{R}$ satisfies $\underline{\lim}\, f_n = 0$ but $\int f_n = 1$ for every n.

Theorem 9.3 – Dominated Convergence (Lebesgue)

Let (f_n) be a sequence of measurable functions with $f_n \to f$ a.e., and suppose there is an integrable function g with $|f_n| \le g$ a.e. for every n. Then f is integrable and

$$\int_X f\, d\mu = \lim_{n\to\infty} \int_X f_n\, d\mu.$$

Moreover, $\int |f_n - f| d\mu \to 0$.

Proof. Passing to a set of full measure, one may suppose convergence and domination at every point. Since $|f| \le g$, f is integrable. Apply Fatou to the two nonnegative sequences $g + f_n \ge 0$ and $g - f_n \ge 0$:

For $g + f_n$:

$$\int g + \int f \le \underline{\lim} \int (g + f_n) = \int g + \underline{\lim} \int f_n,$$

so that $\int f \le \underline{\lim} \int f_n$.

For $g - f_n$:

$$\int g - \int f \le \underline{\lim} \int (g - f_n) = \int g - \overline{\lim} \int f_n,$$

so that $\overline{\lim} \int f_n \le \int f$.

Combining, $\overline{\lim} \int f_n \le \int f \le \underline{\lim} \int f_n$, which gives that the limit exists and equals $\int f$. The claim $\int |f_n - f| d\mu \to 0$ follows by applying DCT to $|f_n - f|$ with dominating function $2g$. □

Corollary 9.1 – Bounded convergence

Let $\mu(X) < \infty$, $f_n \to f$ a.e. and $|f_n| \le M$ a.e. for every n, with M constant. Then f is integrable and $\int_X f\, d\mu = \lim_n \int_X f_n\, d\mu$.

Proof. The constant function $g \equiv M$ is integrable, because $\int_X M\, d\mu = M\mu(X) < \infty$, and it dominates all the f_n. Dominated convergence is applied. □

Example 9.1 – A limit computed with dominated convergence

Compute

$$L = \lim_{n\to\infty} \int_0^\pi (\sin x)^n \sqrt{x}\, dx.$$

The integrand converges pointwise to 0 on $[0,\pi] \setminus \{\pi/2\}$ (there $0 \le \sin x < 1$, so that $(\sin x)^n \to 0$), whereas at $x = \pi/2$ it equals $\sqrt{\pi/2}$ for every n: the pointwise limit is 0 a.e. The convergence is not uniform (near $\pi/2$ the integrand stays close to $\sqrt{\pi/2}$), so that the classical interchange criterion does not apply. But

$$|(\sin x)^n \sqrt{x}| \le \sqrt{x} =: g(x), \qquad \int_0^\pi \sqrt{x}\, dx = \tfrac{2}{3}\pi^{3/2} < \infty,$$

and by dominated convergence

$$L = \int_0^\pi \lim_n (\sin x)^n \sqrt{x}\, dx = \int_0^\pi 0\, dx = 0.$$

9.1 Implications between types of convergence

There are four types of convergence for sequences of measurable functions: *uniform*, *pointwise* (or a.e.), *in measure* and *in L^p*. The elementary implications are shown, together with two fine classical results: Riesz's theorem (an a.e. subsequence from convergence in measure) and the DCT in measure.

Proposition 9.1 – Convergence in L^p implies convergence in measure

If $f_n \to f$ in L^p with $1 \le p < \infty$, then $f_n \to f$ in measure.

Proof. By Chebyshev applied to $|f_n - f|$,

$$\mu(\{|f_n - f| \ge \epsilon\}) \le \frac{\|f_n - f\|_p^p}{\epsilon^p} \to 0.$$

□

Theorem 9.4 – Riesz: a.e. subsequence

If $f_n \to f$ in measure, there is a subsequence (f_{n_k}) with $f_{n_k} \to f$ almost everywhere.

Proof. Construct a subsequence (f_{n_k}) with

$$\mu\Big(\{|f_{n_k} - f| > 2^{-k}\}\Big) < 2^{-k}.$$

This is possible because convergence in measure gives, for each k, an n_k satisfying that, and the n_k can be chosen strictly increasing.

Let $A_k = \{|f_{n_k} - f| > 2^{-k}\}$ and $N = \overline{\lim}_k A_k$. By Borel-Cantelli I (because $\sum \mu(A_k) < \infty$), $\mu(N) = 0$. If $x \notin N$, x is in at most finitely many A_k, so that for large k, $|f_{n_k}(x) - f(x)| \le 2^{-k} \to 0$. Hence $f_{n_k} \to f$ outside N. □

Remark. In general, convergence in measure does *not* imply a.e. convergence (the "walking waves" mentioned in the section on convergence). But *a subsequence* does converge a.e.

Proposition 9.2 – A.e. convergence implies convergence in measure if $\mu(X) < \infty$

If $\mu(X) < \infty$ and $f_n \to f$ a.e., then $f_n \to f$ in measure.

Proof. By Egoroff, for every $\delta > 0$ there is N with $\mu(N) < \delta$ such that $f_n \to f$ uniformly on N^c. Given $\epsilon > 0$, on N^c there is n_0 such that $|f_n - f| < \epsilon$ for $n \geq n_0$, that is, $\{|f_n - f| \geq \epsilon\} \subseteq N$. Hence $\mu(\{|f_n - f| \geq \epsilon\}) \leq \delta$, and δ was arbitrary. □

Theorem 9.5 – Dominated convergence in measure

Let (f_n) be a measurable sequence with $f_n \to f$ in measure, and suppose there is $g \geq 0$ integrable with $|f_n| \leq g$ a.e. for every n. Then $f \in L^1$, $\int |f_n - f| d\mu \to 0$, and in particular $\int f_n d\mu \to \int f \, d\mu$.
More generally, if $|f_n|^p \leq g$ with g integrable and $1 \leq p < \infty$, then $f_n \to f$ in L^p.

Proof. By Riesz's theorem there is a subsequence $f_{n_k} \to f$ a.e., so that $|f| \leq g$ a.e. and $f \in L^1$.

For convergence in L^1, suppose by contradiction that $\int |f_n - f| d\mu \not\to 0$. There is $\epsilon_0 > 0$ and a subsequence (f_{n_j}) with $\int |f_{n_j} - f| d\mu \geq \epsilon_0$. Since $f_{n_j} \to f$ in measure, a sub-subsequence $(f_{n_{j_l}})$ with $f_{n_{j_l}} \to f$ a.e. is extracted. By the classical DCT, $\int |f_{n_{j_l}} - f| d\mu \to 0$, which contradicts the lower bound ϵ_0.

The L^p version is analogous, applying DCT to $|f_n - f|^p \leq 2^p(|f_n|^p + |f|^p) \leq 2^p(g + |f|^p)$. □

Remark. Collecting the implications, in finite measure one has

$$\text{uniform} \implies \text{a.e.} \implies \text{in measure}, \qquad L^p \implies \text{in measure}.$$

And with integrable domination, "in measure" or "a.e." imply "in L^p". The other implications are false in general.

9.2 Density of simple and continuous functions in L^p

Theorem 9.6 – Simple functions are dense in L^p

Let $1 \leq p < \infty$. The set of integrable simple functions (that is, simple φ with $\|\varphi\|_p < \infty$) is dense in $L^p(\mu)$.

Proof. Let $f \in L^p$. Decompose $f = f^+ - f^-$. For $f^+ \geq 0$, by the theorem of approximation by simple functions there are $\varphi_n \nearrow f^+$ with φ_n nonnegative simple. Since $0 \leq \varphi_n \leq f^+$, $\varphi_n \in L^p$. Moreover $|\varphi_n - f^+|^p \leq (f^+)^p \in L^1$, and it converges to 0 pointwise, so that by DCT $\|\varphi_n - f^+\|_p \to 0$. Analogously for f^-. □

Theorem 9.7 – Continuous functions are dense in $L^p(\mathbb{R}, \lambda)$

Let $1 \leq p < \infty$. The continuous functions with compact support $C_c(\mathbb{R})$ are dense in $L^p(\mathbb{R}, \lambda)$.

Proof. By the previous theorem it suffices to approximate integrable simple functions by continuous ones. By linearity, the case $f = \mathbf{1}_A$ with $\lambda(A) < \infty$ suffices.

Reduction to intervals. By construction of the Lebesgue measure (via outer measure with covers by intervals), given $\epsilon > 0$ there is a countable union of open intervals $U = \bigcup_n (a_n, b_n)$ with $A \subseteq U$ and $\lambda(U \setminus A) < \epsilon^p/2$. Truncating to a finite union $V = (a_1, b_1) \cup \cdots \cup (a_N, b_N)$, one may suppose $\lambda(U \setminus V) < \epsilon^p/2$. Then $\|\mathbf{1}_A - \mathbf{1}_V\|_p^p \leq \lambda(A \triangle V) \leq \lambda(U \setminus A) + \lambda(U \setminus V) < \epsilon^p$.

Approximation by continuous functions on each interval. For $\mathbf{1}_{(a,b)}$, define $g_\eta : \mathbb{R} \to [0, 1]$ continuous with support in $(a - \eta, b + \eta)$, equal to 1 on $[a, b]$ and decreasing linearly to 0 at the ends. Then $|\mathbf{1}_{(a,b)} - g_\eta|$ vanishes outside $[a - \eta, a] \cup [b, b + \eta]$ and is bounded by 1, so that

$$\|\mathbf{1}_{(a,b)} - g_\eta\|_p^p \leq 2\eta \to 0 \quad \text{as } \eta \to 0.$$

Combining, $\|\mathbf{1}_V - \sum_k g_\eta^{(k)}\|_p$ becomes arbitrarily small, and the finite sum of continuous functions with compact support is continuous with compact support. Hence for every $\epsilon > 0$ there is $g \in C_c(\mathbb{R})$ with $\|\mathbf{1}_A - g\|_p < 2\epsilon$. □

Remark. Density allows proving properties in L^p by verifying them first on simple (or continuous) functions and then passing to the limit. It is very useful for extending operators: if an operator is linear and continuous on a dense subset, it extends uniquely to the whole space.

10 The L^p spaces

The theory of the L^p spaces as normed spaces (the Young, Hölder and Minkowski inequalities, completeness (Riesz-Fischer theorem), the inclusions between the L^p and the ℓ^p, the parallelogram law, the dual space and the Riesz representation theorem) is developed, with proofs, in the *Real analysis* notes. To avoid duplicating it, only the definition and the statements that the rest of these notes uses are fixed here, and what does not appear there is proved: the orthogonal decomposition of a Hilbert space, on which conditional expectation and the proof of Radon-Nikodym rely.

Recall that, for $1 \le p < \infty$,

$$L^p(\mu) = \left\{ f : X \to \mathbb{R} \text{ measurable} : \int_X |f|^p \, d\mu < \infty \right\}, \qquad \|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p},$$

identifying two functions that coincide a.e. (without that identification, $\|\cdot\|_p$ is only a seminorm: $\|f\|_p = 0$ is equivalent to $f = 0$ a.e.). The facts that will be used, all proved in the *Real analysis* notes:

(i) *Hölder's inequality:* if $\frac{1}{p} + \frac{1}{q} = 1$ with $1 < p, q < \infty$, $f \in L^p(\mu)$ and $g \in L^q(\mu)$, then $fg \in L^1(\mu)$ and $\|fg\|_1 \le \|f\|_p \, \|g\|_q$. For $p = q = 2$ it is the Cauchy-Schwarz inequality.

(ii) *Minkowski's inequality:* $\|f + g\|_p \le \|f\|_p + \|g\|_p$. With it, $(L^p(\mu), \|\cdot\|_p)$ is a normed space.

(iii) *Riesz-Fischer theorem:* $L^p(\mu)$ is complete, that is, a Banach space. (The proof runs on the tools of these notes: monotone convergence, Fatou's Lemma and dominated convergence.)

(iv) $L^2(\mu)$ is a *Hilbert space*: its norm comes from the inner product

$$\langle f, g \rangle = \int_X f g \, d\mu,$$

finite by Cauchy-Schwarz.

(v) *Projection onto a closed convex set:* if H is a Hilbert space, $S \subseteq H$ is convex, closed and nonempty, and $x \in H$, there is a unique $s_0 \in S$ that realizes the distance: $\|x - s_0\| = \inf\{\|x - s\| : s \in S\}$.

(vi) *Riesz representation:* every continuous linear functional $\varphi : H \to \mathbb{R}$ on a Hilbert space is written $\varphi(x) = \langle x, z \rangle$ for a unique $z \in H$. In $L^2(\mu)$: every continuous linear functional is "integrating against a function". This is exactly the form in which the proof of Radon-Nikodym will use it.

Remark. If $\mathcal{G} \subseteq \mathcal{F}$ is a sub-σ-algebra, $L^p(\mathcal{G}) \subseteq L^p(\mathcal{F})$ is a subspace and, being itself complete (Riesz-Fischer), it is a *closed* subspace. Conditional expectation will use this fact.

Theorem 10.1 – Orthogonal decomposition

Let S be a closed subspace of a Hilbert space H, and let $S^\perp = \{v \in H : \langle v, s \rangle = 0 \text{ for every } s \in S\}$. Then $H = S \oplus S^\perp$: every $x \in H$ is written uniquely as $x = s_0 + v$ with $s_0 \in S$ and $v \in S^\perp$.

Proof. A subspace is convex and, by the projection onto a closed convex set (item (v) above), there is the $s_0 \in S$ that realizes the distance from x to S. It is claimed that $x - s_0 \in S^\perp$. Let $s \in S$ and $t \in \mathbb{R}$; since $s_0 + ts \in S$,

$$\|x - s_0\|^2 \le \|x - s_0 - ts\|^2 = \|x - s_0\|^2 - 2t\langle x - s_0, s \rangle + t^2 \|s\|^2.$$

The function $\phi(t) = t^2\|s\|^2 - 2t\langle x - s_0, s\rangle$ is then ≥ 0 for every t and vanishes at $t = 0$: its derivative at 0 must vanish, that is $\langle x - s_0, s\rangle = 0$. Since $s \in S$ was arbitrary, $x - s_0 \in S^\perp$ and $x = s_0 + (x - s_0) \in S + S^\perp$.

The decomposition is unique because $S \cap S^\perp = \{0\}$: if w belongs to both, $\langle w, w\rangle = 0$ and $w = 0$. □

The operator $P : H \to S$, $P(x) = s_0$, is called the *orthogonal projection* onto S.

11 Product measures: Fubini-Tonelli

Given two measures, how does one integrate a function on the product? The theorems of Tonelli and Fubini allow interchanging the order of integration.

Definition 11.1 – Product measure

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. On the product $X \times Y$ define:

- The product σ-algebra $\mathcal{A} \otimes \mathcal{B} = \sigma(\{A \times B : A \in \mathcal{A}, B \in \mathcal{B}\})$.
- The product measure $\mu \otimes \nu$ is the unique measure on $\mathcal{A} \otimes \mathcal{B}$ with $(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$ for measurable rectangles.

Remark. In particular, $\lambda \otimes \lambda$ on $\mathbb{R}^2$ extends to the n-dimensional Lebesgue measure, which assigns to a rectangle $[a, b] \times [c, d]$ the area $(b - a)(d - c)$.

Theorem 11.1 – Tonelli (nonnegative functions)

Let $f : X \times Y \to [0, \infty]$ be measurable with respect to $\mathcal{A} \otimes \mathcal{B}$, with μ and ν σ-finite. Then:

(i) For each $x \in X$, $y \mapsto f(x, y)$ is $\mathcal{B}$-measurable (and symmetrically for y).

(ii) $x \mapsto \int_Y f(x, y)\, d\nu(y)$ is $\mathcal{A}$-measurable (and symmetrically).

(iii) The equality

$$\int_{X \times Y} f \, d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y)\, d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y)\, d\mu(x) \right) d\nu(y)$$

holds.

Proof. *Step 1: measurability of the sections.* Let $\mathcal{M} \subseteq \mathcal{A} \otimes \mathcal{B}$ be the class of sets E such that the section $E_x = \{y : (x, y) \in E\}$ is $\mathcal{B}$-measurable for every x. If $E = A \times B$ is a measurable rectangle, $E_x = B$ if $x \in A$, $\emptyset$ otherwise, in both cases measurable. $\mathcal{M}$ is a σ-algebra (directly: $(E^c)_x = (E_x)^c$, $(\bigcup E_n)_x = \bigcup (E_n)_x$). Since it contains the rectangles, $\mathcal{M} \supseteq \sigma(\text{rectangles}) = \mathcal{A} \otimes \mathcal{B}$. By approximation by simple functions, for measurable f so is $f(x, \cdot)$.

Step 2: case $f = \mathbf{1}_E$, $E \in \mathcal{A} \otimes \mathcal{B}$. Define $g_E(x) = \nu(E_x)$. It must be seen that g_E is $\mathcal{A}$-measurable and $\int_X g_E \, d\mu = (\mu \otimes \nu)(E)$. For rectangles $E = A \times B$, $g_E(x) = \nu(B)\mathbf{1}_A(x)$, measurable, and $\int_X g_E \, d\mu = \mu(A)\nu(B) = (\mu \otimes \nu)(E)$.

The class $\mathcal{D}$ of the E where this property holds is a *monotone class*: closed under increasing unions and decreasing intersections (assuming σ-finiteness in order to interchange limits with integrals via MCT) and under differences of comparable sets. By the monotone class lemma (or by Dynkin's π-λ lemma), $\mathcal{D}$ contains the σ-algebra generated by the rectangles, which is $\mathcal{A} \otimes \mathcal{B}$.

Step 3: case f nonnegative simple. By linearity over the terms $c_i \mathbf{1}_{E_i}$.

Step 4: case $f \geq 0$ general. Take nonnegative simple $\varphi_n \nearrow f$ (approximation by simple functions). For each x, $\varphi_n(x, \cdot) \nearrow f(x, \cdot)$. By MCT on Y,

$$\int_Y \varphi_n(x, y)\, d\nu(y) \nearrow \int_Y f(x, y)\, d\nu(y) =: F(x).$$

F is a pointwise limit of $\mathcal{A}$-measurable functions (by Step 3), so that F is $\mathcal{A}$-measurable. Applying MCT again on X,

$$\int_X F \, d\mu = \lim_n \int_X \int_Y \varphi_n \, d\nu \, d\mu = \lim_n \int_{X \times Y} \varphi_n \, d(\mu \otimes \nu) = \int_{X \times Y} f \, d(\mu \otimes \nu),$$

using the simple case. Symmetrically for the other order. □

Theorem 11.2 – Fubini (integrable functions)

Let $f : X \times Y \to \mathbb{R}$ be measurable. If f is integrable with respect to $\mu \otimes \nu$ (or if, by Tonelli, $\int_X \int_Y |f|\, d\nu\, d\mu < \infty$), then:

(i) For μ-a.e. x, the function $y \mapsto f(x, y)$ is in $L^1(\nu)$.

(ii) The function $x \mapsto \int_Y f(x, y)\, d\nu(y)$, defined μ-a.e., is in $L^1(\mu)$.

(iii) Tonelli's equality holds for f, with finite integrals.

Proof. Decompose $f = f^+ - f^-$ with $f^\pm \geq 0$ measurable. By Tonelli applied to $|f| = f^+ + f^-$,

$$\int_X \int_Y |f|\, d\nu\, d\mu = \int_{X\times Y} |f|\, d(\mu \otimes \nu) < \infty.$$

For the left-hand side to be finite, $\int_Y |f(x, y)|\, d\nu(y) < \infty$ for μ-a.e. x, that is, $f(x, \cdot) \in L^1(\nu)$ μ-a.e. At those points x define

$$F(x) = \int_Y f(x, y)\, d\nu(y) = \int_Y f^+(x, y)\, d\nu(y) - \int_Y f^-(x, y)\, d\nu(y),$$

which is the difference of two measurable functions that are finite μ-a.e., so that F is $\mathcal{A}$-measurable and, by Tonelli applied to $f^\pm$,

$$\int_X F\, d\mu = \int_{X\times Y} f^+\, d(\mu \otimes \nu) - \int_{X\times Y} f^-\, d(\mu \otimes \nu) = \int_{X\times Y} f\, d(\mu \otimes \nu),$$

and all the integrals are finite because $f \in L^1(\mu \otimes \nu)$. □

Remark. In practice, the workflow is: *first* verify with Tonelli (on $|f|$) that the double integral is finite, *then* apply Fubini to interchange the order with f. Skipping the check gives wrong results (the classical computation $\int_0^1 \int_0^1 (x^2 - y^2)/(x^2 + y^2)^2\, dx\, dy = \pi/4$ vs. $\int \int = -\pi/4$ happens because $|f|$ is not integrable).

12 Complete measure spaces

Definition 12.1 – Complete space

$(X, \mathcal{A}, \mu)$ is *complete* if every subset of a set of measure zero is measurable (in particular, it has measure zero):

$$E \in \mathcal{A},\ \mu(E) = 0,\ F \subseteq E \implies F \in \mathcal{A}.$$

Remark. Every measure space can be *completed*: given $(X, \mathcal{A}, \mu)$, define $\overline{\mathcal{A}} = \{A \cup N : A \in \mathcal{A}, N \subseteq E \text{ with } \mu(E) = 0\}$ and extend μ to $\overline{\mathcal{A}}$. For example, $(\mathbb{R}, \mathcal{B}, \lambda)$ is not complete, but its completion is $(\mathbb{R}, \mathcal{L}, \lambda)$.

Remark. In complete spaces, if f is measurable and $g = f$ a.e., then g is also measurable. This makes it easier to handle "a.e." properties.

13 Signed measures and the Radon-Nikodym theorem

Definition 13.1 – Signed measure

A function $\nu : \mathcal{A} \to [-\infty, +\infty]$ is a *signed measure* if:

(i) $\nu(\emptyset) = 0$.

(ii) ν takes at most one of the values $\pm\infty$.

(iii) σ-additivity: $\nu(\bigsqcup A_n) = \sum \nu(A_n)$ (with absolute convergence when the left-hand side is finite).

Definition 13.2 – Absolute continuity

Let μ, ν be measures on $(X, \mathcal{A})$. ν is said to be *absolutely continuous* with respect to μ (written $\nu \ll \mu$) if

$$\mu(E) = 0 \implies \nu(E) = 0 \qquad \text{for every } E \in \mathcal{A}.$$

Definition 13.3 – Mutually singular

μ and ν are *mutually singular* (written $\mu \perp \nu$) if there are disjoint $A, B \in \mathcal{A}$ with $A \cup B = X$ and $\mu(B) = 0 = \nu(A)$.

Example 13.1 – Absolutely continuous measures via integration

If $f \geq 0$ is measurable and μ is a measure, $\nu(A) = \int_A f \, d\mu$ is a measure on the same measurable space, and it is absolutely continuous with respect to μ: if $\mu(A) = 0$, then $\int_A f \, d\mu = 0$ (the integral over a μ-null set is zero). The Radon-Nikodym theorem, later on, will say that *every* absolutely continuous measure (under reasonable hypotheses) has this form.

13.1 Lebesgue decomposition + Radon-Nikodym

The theorem proved next is the combined version: *every* measure ν decomposes into an absolutely continuous part and a singular part with respect to μ. When $\nu \ll \mu$, the singular part disappears and only the derivative remains.

Theorem 13.1 – Lebesgue-Radon-Nikodym

Let μ and ν be σ-finite measures on $(X, \mathcal{A})$. Then:

(i) There are unique measures $\nu_a, \nu_\perp$ such that

$$\nu = \nu_a + \nu_\perp, \qquad \nu_a \ll \mu, \qquad \nu_\perp \perp \mu.$$

(ii) There is a measurable function $h : X \to [0, \infty)$, unique μ-a.e., such that

$$\nu_a(E) = \int_E h \, d\mu \qquad \forall E \in \mathcal{A}.$$

The function h is called the *Radon-Nikodym derivative* of ν_a with respect to μ, and it is denoted $h = \dfrac{d\nu_a}{d\mu}$. In particular, when $\nu \ll \mu$, $\nu_\perp = 0$ and $\nu(E) = \int_E h \, d\mu$.

Proof (von Neumann). The case of finite measures is done; the passage to σ-finite ones is by decomposition into finite parts.

Auxiliary functional. On $L^2(\mu+\nu)$ define

$$\varphi(f) = \int_X f \, d\nu.$$

It is linear and continuous: by Cauchy-Schwarz (with $g \equiv 1$),

$$|\varphi(f)| \le \int_X |f| \, d\nu \le \nu(X)^{1/2} \, \|f\|_{L^2(\nu)} \le \nu(X)^{1/2} \, \|f\|_{L^2(\mu+\nu)}.$$

By the Riesz representation theorem applied to $L^2(\mu+\nu)$, there is $g \in L^2(\mu+\nu)$ such that

$$\int_X f \, d\nu = \int_X f g \, d(\mu+\nu) \qquad \forall f \in L^2(\mu+\nu). \tag{1}$$

$0 \le g \le 1$ *almost everywhere.* Taking $f = \mathbf{1}_A$ in (1),

$$\nu(A) = \int_A g \, d(\mu+\nu),$$

whence $0 \le \int_A g \, d(\mu+\nu) \le (\mu+\nu)(A)$. This forces $0 \le g \le 1$ $(\mu+\nu)$-a.e.

Decomposition. Let $N = \{x : g(x) = 1\}$ and $M = N^c = \{x : g(x) < 1\}$. Define

$$\nu_\perp(E) = \nu(E \cap N), \qquad \nu_a(E) = \nu(E \cap M).$$

It is proved that $\nu_\perp \perp \mu$ and $\nu_a \ll \mu$:

- $\mu(N) = 0$: taking $f = \mathbf{1}_N$ in (1), $\nu(N) = \int_N g \, d(\mu+\nu) = (\mu+\nu)(N) = \mu(N) + \nu(N)$, whence $\mu(N) = 0$. Thus $\nu_\perp$ and μ are mutually singular (separated by N).
- If $\mu(E) = 0$, it must hold that $\nu_a(E) = \nu(E \cap M) = 0$. Rearranging (1) (for $f = \mathbf{1}_{E\cap M}$),

  $$\int_{E\cap M} (1-g) \, d\nu = \int_{E\cap M} g \, d\mu = 0.$$

  On M, $1 - g > 0$, so that necessarily $\nu(E \cap M) = 0$.

Derivative. Iterating (1) with $f_n = \mathbf{1}_E(1 + g + g^2 + \cdots + g^n)$ and subtracting, one obtains

$$\int_E (1 - g^{n+1}) \, d\nu = \int_E (g + g^2 + \cdots + g^{n+1}) \, d\mu.$$

On M, $g^{n+1} \to 0$ and $g + g^2 + \cdots \to g/(1-g)$. By monotone convergence,

$$\nu_a(E) = \nu(E \cap M) = \int_{E\cap M} \frac{g}{1-g} \, d\mu = \int_E h \, d\mu,$$

where $h = \mathbf{1}_M \cdot g/(1-g)$.

Uniqueness follows in the standard way: if two decompositions coexisted, their difference would be at the same time absolutely continuous and singular, which forces it to be zero. □

13.2 Application: the Fundamental Theorem of Calculus

Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing and right-continuous, and λ_F its Lebesgue-Stieltjes measure. If $\lambda_F \ll \lambda$ (which happens, for example, when F is *absolutely continuous* in the classical sense), by Radon-Nikodym there is $f \in L^1_{\text{loc}}(\mathbb{R})$ such that

$$F(b) - F(a) = \lambda_F((a, b]) = \int_{(a,b]} f \, d\lambda = \int_a^b f(x) \, dx.$$

The function $f = d\lambda_F/d\lambda$ plays the role of the derivative of F in the distributional sense. This is the *general* version of the Fundamental Theorem of Calculus.

14 Conditional Expectation

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Recall elementary conditional probability: given events A, B with $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This is a "basic" version that only allows conditioning on an event. What is sought now is to condition with respect to an *entire* σ*-algebra* $\mathcal{G} \subseteq \mathcal{F}$, which represents the available information.

Remark. Two direct consequences of the elementary definition. If A and B are independent ($P(A \cap B) = P(A)P(B)$), then $P(A \mid B) = P(A)$: knowing that B occurred provides no information about A. And if $\{B_1, B_2, \dots\}$ is a partition of Ω with $P(B_i) > 0$, the sets $\{A \cap B_i\}$ partition A, so that

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A \mid B_i)\, P(B_i) \qquad \text{(total probability)},$$

and solving from $P(B_k \mid A)\, P(A) = P(A \cap B_k) = P(A \mid B_k)\, P(B_k)$ one obtains **Bayes' formula**:

$$P(B_k \mid A) = \frac{P(A \mid B_k)\, P(B_k)}{\sum_i P(A \mid B_i)\, P(B_i)}.$$

Conditional expectation with respect to a σ-algebra will generalize this weighted average over the elements of a partition.

14.1 Motivation

Example 14.1 – Three coin tosses

A coin is tossed 3 times. Ω has 8 elements $\{HHH, HHT, \dots, TTT\}$, and $\mathcal{F} = \mathcal{P}(\Omega)$. Define nested σ-algebras that "gradually reveal" information:

$$\begin{aligned}
&\widetilde{\mathcal{F}}_0 = \{\emptyset, \Omega\} && \text{(nothing is known)},\\
&\widetilde{\mathcal{F}}_1 = \sigma(A) = \{\emptyset, \Omega, A, A^c\} && \text{where } A = \{\text{the 1st toss is } H\},\\
&\widetilde{\mathcal{F}}_2 = \sigma(B_1, B_2, B_3, B_4) && \text{(according to the first 2 tosses)},\\
&\widetilde{\mathcal{F}}_3 = \mathcal{P}(\Omega) && \text{(everything is known)}.
\end{aligned}$$

Each $\widetilde{\mathcal{F}}_k$ has more information than $\widetilde{\mathcal{F}}_{k-1}$.

Definition 14.1 – Filtration

A *filtration* on $(\Omega, \mathcal{F}, P)$ is an increasing family of σ-algebras $(\widetilde{\mathcal{F}}_t)_{t \in T} \subseteq \mathcal{F}$:

$$s \le t \implies \widetilde{\mathcal{F}}_s \subseteq \widetilde{\mathcal{F}}_t.$$

Example 14.2 – Conditional expectation in the three tosses

Continuing the previous example, let $X : \Omega \to \mathbb{R}$ with $X(\omega) =$ "number of heads in ω". The expectation is

$$E(X) = 0 \cdot \tfrac{1}{8} + 1 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 3 \cdot \tfrac{1}{8} = \tfrac{3}{2}.$$

If the 1st toss has already been seen (that is, $\widetilde{\mathcal{F}}_1$ is known), one *expects* on average:

$$E(X \mid \widetilde{\mathcal{F}}_1)(\omega) = \begin{cases} 2, & \text{if the 1st was } H \ (= 1 + E(\text{2 remaining tosses})), \\ 1, & \text{if the 1st was } T. \end{cases}$$

Note that $E(X \mid \widetilde{\mathcal{F}}_1)$ is an $\widetilde{\mathcal{F}}_1$-measurable *random variable* (it takes constant values on A and on A^c).

Remark. In the example measurability can be verified by hand: $\{E(X \mid \widetilde{\mathcal{F}}_1) \geq 2\} = A \in \widetilde{\mathcal{F}}_1$, whereas $\{X \geq 2\} = \{HHH, HHT, HTH, THH\} \notin \widetilde{\mathcal{F}}_1$. The conditional expectation is $\widetilde{\mathcal{F}}_1$-measurable; X is not.

Remark. (*Definition on partitions.*) When $\mathcal{G} = \sigma(G_1, G_2, \dots)$ with $\{G_i\}$ a partition of Ω and $P(G_i) > 0$, there is an explicit formula. A $\mathcal{G}$-measurable variable is constant on each atom G_i, and the conditional expectation turns out to be the average of X over the atom where ω falls:

$$E(X \mid \mathcal{G})(\omega) = \frac{1}{P(G_i)} \int_{G_i} X \, dP \qquad \text{if } \omega \in G_i.$$

The characteristic property is checked: every $A \in \mathcal{G}$ is a union of atoms, $A = \bigcup_{i \in M} G_i$, and

$$\int_A E(X \mid \mathcal{G}) \, dP = \sum_{i \in M} \left(\frac{1}{P(G_i)} \int_{G_i} X \, dP \right) P(G_i) = \sum_{i \in M} \int_{G_i} X \, dP = \int_A X \, dP.$$

The general definition (which follows) extends this to σ-algebras without atoms, where the local average can no longer be written by dividing by $P(G_i)$.

14.2 General definition

Definition 14.2 – Conditional expectation

Let $X \in L^1(\Omega, \mathcal{F}, P)$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra. The *conditional expectation* of X given $\mathcal{G}$, denoted $E(X \mid \mathcal{G})$, is the unique $\mathcal{G}$-measurable random variable (up to a.e. equality) such that

$$\int_A E(X \mid \mathcal{G}) \, dP = \int_A X \, dP \qquad \forall A \in \mathcal{G}.$$

Theorem 14.1 – Existence (via Radon-Nikodym)

Under the previous hypotheses, $E(X \mid \mathcal{G})$ exists and is unique P-a.e.

Proof. Suppose first $X \geq 0$. Define $\lambda : \mathcal{G} \to [0, \infty]$ by

$$\lambda(A) = \int_A X \, dP.$$

λ is a measure on $(\Omega, \mathcal{G})$. It is absolutely continuous with respect to $P|_{\mathcal{G}}$: if $P(A) = 0$, then $\int_A X\, dP = 0$. By Radon-Nikodym, there is $h \in L^1(\mathcal{G})$ with

$$\int_A X\, dP = \lambda(A) = \int_A h\, dP \qquad \forall A \in \mathcal{G}.$$

That h is $E(X \mid \mathcal{G})$. For general X, decompose $X = X^+ - X^-$ and subtract. □

14.3 Properties

Proposition 14.1 – Properties of conditional expectation

Let $X, Y \in L^1$ and $\mathcal{G}, \mathcal{H} \subseteq \mathcal{F}$ be σ-algebras.

(i) $E(aX + bY \mid \mathcal{G}) = aE(X \mid \mathcal{G}) + bE(Y \mid \mathcal{G})$.

(ii) If Y is $\mathcal{G}$-measurable (and $XY \in L^1$),

$$E(XY \mid \mathcal{G}) = Y \cdot E(X \mid \mathcal{G}).$$

(iii) If X is independent of $\mathcal{G}$ (that is, $X^{-1}(B) \perp A$ for every $A \in \mathcal{G}$ and Borel set B),

$$E(X \mid \mathcal{G}) = E(X).$$

(iv) If $\mathcal{H} \subseteq \mathcal{G}$,

$$E(E(X \mid \mathcal{G}) \mid \mathcal{H}) = E(X \mid \mathcal{H}).$$

Proof.
(i) For every $A \in \mathcal{G}$,

$$\begin{aligned}\int_A (aE(X \mid \mathcal{G}) + bE(Y \mid \mathcal{G}))\, dP &= a\int_A E(X \mid \mathcal{G})\, dP + b\int_A E(Y \mid \mathcal{G})\, dP \\ &= a\int_A X\, dP + b\int_A Y\, dP = \int_A (aX + bY)\, dP.\end{aligned}$$

By uniqueness, $E(aX + bY \mid \mathcal{G}) = aE(X \mid \mathcal{G}) + bE(Y \mid \mathcal{G})$.

(ii) It is proved by increasing approximation.

- For $Y = \mathbf{1}_C$ with $C \in \mathcal{G}$: for every $A \in \mathcal{G}$, $A \cap C \in \mathcal{G}$, and

  $$\int_A \mathbf{1}_C \cdot E(X \mid \mathcal{G}) dP = \int_{A\cap C} E(X \mid \mathcal{G}) dP = \int_{A\cap C} X dP = \int_A \mathbf{1}_C X dP.$$

  By uniqueness, $E(\mathbf{1}_C X \mid \mathcal{G}) = \mathbf{1}_C E(X \mid \mathcal{G})$.
- For $Y = \sum_i c_i \mathbf{1}_{C_i}$ simple and $\mathcal{G}$-measurable: by linearity from (i).
- For $Y \geq 0$ $\mathcal{G}$-measurable: take nonnegative simple $Y_n \nearrow Y$. If $X \geq 0$, by monotone convergence $\int_A Y_n X dP \to \int_A YX dP$ and $\int_A Y_n E(X \mid \mathcal{G}) dP \to \int_A YE(X \mid \mathcal{G}) dP$. For general X decompose $X = X^+ - X^-$.
- For general Y: $Y = Y^+ - Y^-$ and the previous step is applied to each part.

(iii) For $A \in \mathcal{G}$, $\mathbf{1}_A$ and X are independent, so that $E(\mathbf{1}_A X) = P(A)E(X)$:

$$\int_A X\, dP = E(\mathbf{1}_A X) = E(\mathbf{1}_A)E(X) = P(A)E(X) = \int_A E(X)\, dP.$$

$E(X)$ is constant (in particular $\mathcal{G}$-measurable), so that by uniqueness $E(X \mid \mathcal{G}) = E(X)$.

(iv) Let $Z = E(X \mid \mathcal{G})$ and $W = E(X \mid \mathcal{H})$. For $A \in \mathcal{H} \subseteq \mathcal{G}$,

$$\int_A Z\, dP = \int_A X\, dP = \int_A W\, dP.$$

Since W is $\mathcal{H}$-measurable and the equality holds for every $A \in \mathcal{H}$, by uniqueness of $E(Z \mid \mathcal{H})$ one has $E(Z \mid \mathcal{H}) = W$, that is $E(E(X \mid \mathcal{G}) \mid \mathcal{H}) = E(X \mid \mathcal{H})$. □

14.4 Interpretation as a projection

When $X \in L^2$, conditional expectation admits a geometric characterization:

Theorem 14.2 – $E(\cdot \mid \mathcal{G})$ is an orthogonal projection

For $X \in L^2(\mathcal{F})$, $E(X \mid \mathcal{G})$ is the orthogonal projection of X onto the closed subspace $L^2(\mathcal{G}) \subseteq L^2(\mathcal{F})$. That is, $E(X \mid \mathcal{G})$ is the unique $\mathcal{G}$-measurable element that minimizes

$$\|X - Y\|_{L^2}^2 = E((X-Y)^2) \qquad \text{among } Y \in L^2(\mathcal{G}).$$

Proof. Since $L^2(\mathcal{G})$ is a closed subspace of $L^2(\mathcal{F})$ (Hilbert), there is the projection $\widehat{X}$ characterized by $X - \widehat{X} \perp L^2(\mathcal{G})$, that is

$$\int_\Omega (X - \widehat{X}) Y\, dP = 0 \qquad \forall Y \in L^2(\mathcal{G}).$$

Taking $Y = \mathbf{1}_A$ with $A \in \mathcal{G}$, $\int_A X\, dP = \int_A \widehat{X}\, dP$, and since $\widehat{X}$ is $\mathcal{G}$-measurable, by uniqueness $\widehat{X} = E(X \mid \mathcal{G})$. □

Remark. This explains the name "best predictor of X given the information $\mathcal{G}$": $E(X \mid \mathcal{G})$ minimizes the mean squared error among all $\mathcal{G}$-measurable functions.

Example 14.3 – The three tosses as nested spaces

In the example of the three tosses, $L^2(\widetilde{\mathcal{F}}_k)$ is the space of functions that are constant on the atoms of $\widetilde{\mathcal{F}}_k$; identifying each variable with the vector of its 8 values $(X(HHH), X(HHT), X(HTH), X(HTT), X(THH), X(THT), X(TTH), X(TTT))$,

$$\underbrace{L^2(\widetilde{\mathcal{F}}_0)}_{\dim 1} \subset \underbrace{L^2(\widetilde{\mathcal{F}}_1)}_{\dim 2} \subset \underbrace{L^2(\widetilde{\mathcal{F}}_2)}_{\dim 4} \subset \underbrace{L^2(\widetilde{\mathcal{F}}_3)}_{\dim 8}.$$

The variable $X =$ "number of heads" is the vector $(3,2,2,1,2,1,1,0)$, and the successive orthogonal projections average over increasingly coarse atoms:

$$(3,2,2,1,2,1,1,0) \;\mapsto\; \left(\tfrac{5}{2},\tfrac{5}{2},\tfrac{3}{2},\tfrac{3}{2},\tfrac{3}{2},\tfrac{3}{2},\tfrac{1}{2},\tfrac{1}{2}\right) \;\mapsto\; (2,2,2,2,1,1,1,1) \;\mapsto\; \left(\tfrac{3}{2},\dots,\tfrac{3}{2}\right),$$

which are $E(X \mid \widetilde{\mathcal{F}}_2)$, $E(X \mid \widetilde{\mathcal{F}}_1)$ and $E(X \mid \widetilde{\mathcal{F}}_0) = E(X) = 3/2$. Composing projections $(L^2(\mathcal{F}) \to L^2(\mathcal{G}) \to L^2(\mathcal{H})$ with $\mathcal{H} \subseteq \mathcal{G})$ is the geometric reading of the tower property.

Remark. Another characterization, in terms of covariances: for $X \in L^2$, $E(X \mid \mathcal{G})$ is the unique $\mathcal{G}$-measurable variable with $E(E(X \mid \mathcal{G})) = E(X)$ such that

$$\operatorname{Cov}(E(X \mid \mathcal{G}), Y) = \operatorname{Cov}(X, Y) \qquad \text{for every } Y \in L^2(\mathcal{G}).$$

Indeed, subtracting,

$$\operatorname{Cov}(X, Y) - \operatorname{Cov}(E(X \mid \mathcal{G}), Y) = \big\langle X - E(X \mid \mathcal{G}),\, Y - E(Y) \big\rangle = 0,$$

because $X - E(X \mid \mathcal{G})$ is orthogonal to $L^2(\mathcal{G})$ and $Y - E(Y) \in L^2(\mathcal{G})$. Conditional expectation preserves all the correlation with what is observable.

14.5 Conditional Jensen inequality

Theorem 14.3 – Conditional Jensen

Let $X \in L^1$ and $\varphi : \mathbb{R} \to \mathbb{R}$ convex with $\varphi(X) \in L^1$. Then

$$\varphi(E(X \mid \mathcal{G})) \le E(\varphi(X) \mid \mathcal{G}) \quad P\text{-a.e.}$$

Proof. The idea is the same as in the integral Jensen inequality: by convexity there is a countable family of supporting lines $\varphi(y) = \sup_n (a_n y + b_n)$. For each fixed n, by linearity and monotonicity of conditional expectation,

$$E(\varphi(X) \mid \mathcal{G}) \ge E(a_n X + b_n \mid \mathcal{G}) = a_n E(X \mid \mathcal{G}) + b_n.$$

Taking the supremum over n, $E(\varphi(X) \mid \mathcal{G}) \ge \varphi(E(X \mid \mathcal{G}))$. □

Remark. Particular cases: $\varphi(t) = |t|^p$ with $p \ge 1$ gives

$$|E(X \mid \mathcal{G})|^p \le E(|X|^p \mid \mathcal{G}),$$

and taking expectations, $\|E(X \mid \mathcal{G})\|_p \le \|X\|_p$. Thus, the conditional expectation operator is *contractive* on L^p.

15 Martingales

Martingales formalize the idea of a *fair game*: the conditional expectation of the future value, knowing everything that has happened so far, coincides with the current value.

15.1 Adapted processes and stopping times

Definition 15.1 – Adapted process

Let $(\widetilde{\mathcal{F}}_t)_{t\in T}$ be a filtration. A stochastic process $(X_t)_{t\in T}$ is *adapted* to the filtration if each X_t is $\widetilde{\mathcal{F}}_t$-measurable.

Remark. If no filtration is given, the *natural filtration* $\widetilde{\mathcal{F}}_t = \sigma(X_s : s \le t)$ can be used, which is the smallest σ-algebra that makes $(X_s)_{s\le t}$ measurable. By construction, (X_t) is adapted to its natural filtration.

Definition 15.2 – Stopping time

Let $(\widetilde{\mathcal{F}}_t)_{t\in T}$ be a filtration. A *stopping time* is a random variable $\tau : \Omega \to T$ such that for every t,

$$\{\tau \le t\} \in \widetilde{\mathcal{F}}_t.$$

In discrete time $T = \mathbb{N}$ it is equivalent to requiring $\{\tau = n\} \in \widetilde{\mathcal{F}}_n$.

Remark. Intuitively, τ is an instant *decidable with the information up to that moment.* No knowledge of the future is required.

Example 15.1 – Common stopping times

The following are stopping times: playing a fixed number of times ($\tau \equiv N$ constant), "the first time the event A appears" ($\tau = \inf\{n : \omega_n \in A\}$), or "the k-th visit to A". "The instant before the maximum" is *not* one, because knowing that the next step will be the maximum requires knowing the future. Moreover, $\max(\tau_1, \tau_2)$ and $\min(\tau_1, \tau_2)$ are stopping times: $\{\max(\tau_1, \tau_2) \le t\} = \{\tau_1 \le t\} \cap \{\tau_2 \le t\}$ and $\{\min(\tau_1, \tau_2) \le t\} = \{\tau_1 \le t\} \cup \{\tau_2 \le t\}$ are in $\widetilde{\mathcal{F}}_t$.

Theorem 15.1 – Wald's Lemma

Let (X_i) be i.i.d. and integrable, $S_n = \sum_{i=1}^{n} X_i$, and τ a stopping time with respect to $\widetilde{\mathcal{F}}_n = \sigma(X_1, \dots, X_n)$ with $E(\tau) < \infty$. Then

$$E(S_\tau) = E(X_1)\, E(\tau).$$

Proof. Suppose first $X_i \ge 0$. Write

$$S_\tau = \sum_{n=1}^{\infty} X_n \, \mathbf{1}_{\{\tau \ge n\}}.$$

The event $\{\tau \ge n\} = \{\tau \le n-1\}^c \in \widetilde{\mathcal{F}}_{n-1}$ is independent of X_n, so that $E(X_n \mathbf{1}_{\{\tau\ge n\}}) =$

$E(X_n)\,P(\tau \geq n) = E(X_1)\,P(\tau \geq n)$. By monotone convergence (the summands are nonnegative),

$$E(S_\tau) = \sum_{n=1}^{\infty} E(X_n \mathbf{1}_{\{\tau \geq n\}}) = E(X_1) \sum_{n=1}^{\infty} P(\tau \geq n) = E(X_1)\,E(\tau),$$

using the identity $E(\tau) = \sum_{n\geq 1} P(\tau \geq n)$, valid for variables with values in $\mathbb{N}$ (it is seen by writing $\tau = \sum_{n\geq 1} \mathbf{1}_{\{\tau \geq n\}}$ and interchanging expectation and sum by monotone convergence). For general X_i the above is applied to the parts X_i^+ and X_i^- and subtracted; both expectations are finite since $E|X_1| < \infty$ and $E(\tau) < \infty$. □

Example 15.2 – Compound dice

A die is rolled and N comes up; then N dice are rolled and added: $S = X_1 + \cdots + X_N$. Here N is independent of the X_i (a particular case of stopping time), and

$$E(S) = E(X_1)\,E(N) = \frac{7}{2} \cdot \frac{7}{2} = \frac{49}{4} = 12.25,$$

with no need to condition by hand on the 6 values of N.

15.2 Definition of a martingale

Definition 15.3 – Martingale

A process $(X_n)_{n\in\mathbb{N}}$ adapted to $(\widetilde{\mathcal{F}}_n)$ is a *martingale* if each X_n is integrable and

$$E(X_n \mid \widetilde{\mathcal{F}}_{n-1}) = X_{n-1} \qquad \forall n \geq 1.$$

Remark. Equivalently, $E(X_n \mid \widetilde{\mathcal{F}}_m) = X_m$ for every $m \leq n$ (by the tower property).

Remark. As a consequence, $E(X_n) = E(X_0)$ for every n. All the variables of a martingale have the same expectation.

Definition 15.4 – Submartingale and supermartingale

An adapted integrable process (X_n) is a *submartingale* if

$$E(X_n \mid \widetilde{\mathcal{F}}_{n-1}) \geq X_{n-1} \qquad \forall n \geq 1,$$

and a *supermartingale* if the opposite inequality holds, $E(X_n \mid \widetilde{\mathcal{F}}_{n-1}) \leq X_{n-1}$.

Remark. In the language of games: a martingale is a fair game, a submartingale one favorable to the player and a supermartingale an unfavorable one. A process is a martingale if and only if it is both a sub- and a supermartingale, and the expectations $E(X_n)$ turn out to be increasing in the sub case and decreasing in the super case. If (X_n) is a martingale and φ is convex with $\varphi(X_n)$ integrable, then $(\varphi(X_n))$ is a submartingale: by conditional Jensen (Theorem 14.3), $E(\varphi(X_n) \mid \widetilde{\mathcal{F}}_{n-1}) \geq \varphi(E(X_n \mid \widetilde{\mathcal{F}}_{n-1})) = \varphi(X_{n-1})$. In particular $(|X_n|)$ and $(|X_n|^p)$, with $p \geq 1$, are submartingales.

Example 15.3 – Constants and averages

If $X \in L^1$ and one defines $X_n = E(X \mid \widetilde{\mathcal{F}}_n)$, then (X_n) is a martingale with respect to $(\widetilde{\mathcal{F}}_n)$. By the tower property, $E(X_n \mid \widetilde{\mathcal{F}}_{n-1}) = E(E(X \mid \widetilde{\mathcal{F}}_n) \mid \widetilde{\mathcal{F}}_{n-1}) = E(X \mid \widetilde{\mathcal{F}}_{n-1}) = X_{n-1}$.

Example 15.4 – Symmetric random walk

Let $(X_n)_{n\geq 1}$ be i.i.d. with $E(X_n) = 0$ and $S_n = \sum_{k=1}^{n} X_k$ (with $S_0 = 0$). With respect to the natural filtration,

$$E(S_n \mid \widetilde{\mathcal{F}}_{n-1}) = E(S_{n-1} + X_n \mid \widetilde{\mathcal{F}}_{n-1}) = S_{n-1} + E(X_n) = S_{n-1},$$

using that S_{n-1} is $\widetilde{\mathcal{F}}_{n-1}$-measurable and X_n is independent. Thus (S_n) is a martingale.

Example 15.5 – $S_n^2 - n$

With X_k Rademacher (± 1 with probability $1/2$), $(S_n^2 - n)$ is a martingale. Computation:

$$\begin{aligned} E(S_n^2 \mid \widetilde{\mathcal{F}}_{n-1}) &= E((S_{n-1} + X_n)^2 \mid \widetilde{\mathcal{F}}_{n-1}) = S_{n-1}^2 + 2S_{n-1}E(X_n) + E(X_n^2) \\ &= S_{n-1}^2 + 0 + 1. \end{aligned}$$

Subtracting n from both sides, $E(S_n^2 - n \mid \widetilde{\mathcal{F}}_{n-1}) = S_{n-1}^2 - (n-1)$.

15.3 Predictable strategy: martingale transform

Definition 15.5 – Predictable process

$(H_n)_{n\geq 1}$ is *predictable* (with respect to $(\widetilde{\mathcal{F}}_n)$) if each H_n is $\widetilde{\mathcal{F}}_{n-1}$-measurable. Intuitively, H_n is decided *before* step n.

Proposition 15.1 – Martingale transform

If (X_n) is a martingale and (H_n) is predictable and bounded, then

$$Y_n = \sum_{k=1}^{n} H_k(X_k - X_{k-1})$$

is a martingale.

Proof. $Y_n - Y_{n-1} = H_n(X_n - X_{n-1})$. Since H_n is $\widetilde{\mathcal{F}}_{n-1}$-measurable,

$$E(Y_n - Y_{n-1} \mid \widetilde{\mathcal{F}}_{n-1}) = H_n E(X_n - X_{n-1} \mid \widetilde{\mathcal{F}}_{n-1}) = H_n \cdot 0 = 0.$$

□

Remark. The interpretation: in a fair game, no predictable strategy (one that only uses the past) can turn the game into a favorable one. This justifies mathematically that "there is no infallible system" in a fair game.

Remark. A famous strategy: *double or nothing* (the classical martingale, where the name comes from). $H_n = 2^{n-1}$ if $X_{k-1} = -k$ for every $k < n$, and $H_n = 0$ if the player has already won. The accumulated gain is 1 when the player finally wins, but ruin arrives earlier with significant probability.

16 Doob's theorems

16.1 Optional Stopping

Theorem 16.1 – Optional Stopping (bounded version)

Let (X_n) be a martingale and τ a stopping time with $\tau \leq M$ bounded. Then

$$E(X_\tau) = E(X_0).$$

Proof. Define $X_n^\tau = X_{\tau \wedge n}$, the process "stopped" at τ. It is verified that (X_n^τ) is a martingale (transform by the predictable strategy $H_n = \mathbf{1}_{\tau \geq n}$, which is $\widetilde{\mathcal{F}}_{n-1}$-measurable because $\{\tau \geq n\} = \{\tau \leq n-1\}^c$). Thus $E(X_\tau) = E(X_M^\tau) = E(X_0^\tau) = E(X_0)$. □

Theorem 16.2 – Optional Stopping (uniformly integrable martingale)

If (X_n) is a uniformly integrable martingale (there is X_∞ with $X_n \to X_\infty$ in L^1) and τ is a stopping time that is finite a.e., then

$$E(X_\tau) = E(X_0).$$

Proof. Let $\tau_n = \tau \wedge n$, which is a bounded stopping time. By the bounded version, $E(X_{\tau_n}) = E(X_0)$ for every n.

Since $\tau < \infty$ a.e., $X_{\tau_n} \to X_\tau$ a.e. (eventually $\tau_n = \tau$). It remains to see that $\int X_{\tau_n} dP \to \int X_\tau dP$. Write

$$X_{\tau_n} = X_\tau \mathbf{1}_{\tau \leq n} + X_n \mathbf{1}_{\tau > n}.$$

Since (X_n) is uniformly integrable, in particular $\{X_n \mathbf{1}_{\tau > n}\}$ is UI (their absolute values are dominated by those of (X_n)), and since $P(\tau > n) \to 0$, that term tends to zero in L^1. The first term $X_\tau \mathbf{1}_{\tau \leq n}$ converges to X_τ in L^1 by dominated convergence (with dominating function $|X_\tau| \in L^1$, which also follows from UI: by Fatou and UI, $E|X_\tau| \leq \underline{\lim}\, E|X_{\tau_n}| < \infty$).

Letting $n \to \infty$, $E(X_0) = E(X_{\tau_n}) \to E(X_\tau)$. □

16.2 Doob's inequality

Theorem 16.3 – Doob's maximal inequality

Let (X_n) be a martingale with $X_n \geq 0$, and $c > 0$. Then

$$P\left(\max_{k \leq n} X_k \geq c\right) \leq \frac{E(X_n)}{c}.$$

Proof. Define the stopping time

$$\tau = \min\{m : \max_{k \leq m} X_k \geq c\} \wedge n.$$

By construction $\tau \leq n$ and $\{\tau < n\} = \{\max_k X_k \geq c\}$ (if the maximum reaches c before n). By Optional Stopping,

$$E(X_n) = E(X_\tau) = E(X_\tau \mathbf{1}_{\tau < n}) + E(X_\tau \mathbf{1}_{\tau = n}) \geq c \cdot P(\tau < n),$$

because on $\{\tau < n\}$ one has $X_\tau \geq c$ and $X_\tau \mathbf{1}_{\tau=n} \geq 0$. Solving, $P(\max X_k \geq c) = P(\tau < n) \leq E(X_n)/c$. □

Corollary 16.1 – Gaussian tail of the supremum of Brownian motion

Let (W_t) be a Brownian motion. For all $a, T > 0$,

$$P\left(\sup_{t\leq T} W_t \geq a\right) \leq e^{-a^2/(2T)}.$$

Proof. Fix $\lambda > 0$. If $Z \sim \mathcal{N}(0,t)$, completing the square one computes $E(e^{\lambda Z}) = e^{\lambda^2 t/2}$, so that the process

$$M_t = e^{\lambda W_t - \lambda^2 t/2}$$

is a positive martingale: for $s < t$, since $W_t - W_s \sim \mathcal{N}(0, t-s)$ is independent of the past,

$$E(M_t \mid \mathcal{F}_s) = M_s\, e^{-\lambda^2(t-s)/2}\, E\big(e^{\lambda(W_t - W_s)}\big) = M_s\, e^{-\lambda^2(t-s)/2}\, e^{\lambda^2(t-s)/2} = M_s.$$

Since e^x is increasing and $e^{-\lambda^2 t/2} \geq e^{-\lambda^2 T/2}$ for $t \leq T$, the inclusion of events

$$\left\{\sup_{t\leq T} W_t \geq a\right\} \subseteq \left\{\sup_{t\leq T} M_t \geq e^{\lambda a - \lambda^2 T/2}\right\}$$

holds, and Doob's maximal inequality applied to the positive martingale (M_t) (the continuous version is obtained from the discrete one by taking suprema over dyadic times and using the continuity of the paths), with $E(M_T) = E(M_0) = 1$, gives

$$P\left(\sup_{t\leq T} W_t \geq a\right) \leq \frac{E(M_T)}{e^{\lambda a - \lambda^2 T/2}} = e^{\frac{\lambda^2 T}{2} - \lambda a}.$$

The right-hand side is minimized at $\lambda = a/T$ (setting the derivative $\lambda T - a$ of the exponent to zero), and substituting yields $e^{a^2/(2T) - a^2/T} = e^{-a^2/(2T)}$. □

16.3 Convergence of martingales

Theorem 16.4 – Martingale convergence (Doob)

Let (X_n) be a martingale with $\sup_n E|X_n| < \infty$ (bounded in L^1). Then there is $X \in L^1$ with $X_n \to X$ almost everywhere.

Proof. *Upcrossings* technique. Given $a < b$, define the number of complete upcrossings $U_n^{[a,b]}(\omega)$ as the number of times that $(X_k(\omega))_{k=0}^n$ crosses from below a to above b. More precisely, let $\tau_0 = 0$ and, recursively,

$$\tau_{2j+1} = \min\{k > \tau_{2j} : X_k < a\}, \qquad \tau_{2j+2} = \min\{k > \tau_{2j+1} : X_k > b\},$$

with the convention $\min \emptyset = \infty$. Then $U_n^{[a,b]} = \max\{j : \tau_{2j} \leq n\}$.

Doob's upcrossing inequality. For any martingale (X_n),

$$(b-a)\, E(U_n^{[a,b]}) \leq E((X_n - a)^+).$$

Idea of the proof: the *strategy* $H_k = \mathbf{1}_{\tau_{2j+1}<k\le\tau_{2j+2} \text{ for some } j}$ (buy after falling below a, sell after rising above b) is predictable and bounded. The accumulated gain $Y_n = \sum_{k=1}^n H_k(X_k - X_{k-1})$ is nonnegative because each complete upcrossing from a to b contributes at least $b-a$; more formally, $Y_n \ge (b-a)U_n^{[a,b]} - (X_n - a)^+$ (the remaining term absorbs a possible incomplete upcrossing at the end). Since Y_n is a martingale (transform by a predictable process), $E(Y_n) = 0$, and one gets $(b-a)E(U_n) \le E((X_n-a)^+)$.

Conclusion. If $\sup_n E|X_n| < \infty$, then $E((X_n-a)^+) \le |a| + E|X_n|$ is uniformly bounded, so that

$$E(U_\infty^{[a,b]}) = \lim_n E(U_n^{[a,b]}) \le \frac{|a| + \sup_n E|X_n|}{b-a} < \infty.$$

In particular, $U_\infty^{[a,b]} < \infty$ almost everywhere.

Let $\Lambda_{a,b} = \{\omega : \underline{\lim}\, X_n(\omega) < a < b < \overline{\lim}\, X_n(\omega)\}$. On $\Lambda_{a,b}$, the sequence $X_n(\omega)$ crosses the interval $[a,b]$ infinitely many times, so that $U_\infty^{[a,b]}(\omega) = \infty$. Hence $P(\Lambda_{a,b}) = 0$.

Taking a countable union over rationals $a < b$,

$$P\left(\bigcup_{a<b\in\mathbb{Q}} \Lambda_{a,b}\right) = 0.$$

Outside that null set, $\underline{\lim}\, X_n = \overline{\lim}\, X_n$, that is, the limit $X = \lim X_n$ exists in $\mathbb{R}\cup\{\pm\infty\}$. By Fatou's Lemma applied to $|X_n|$, $E|X| \le \underline{\lim}\, E|X_n| \le \sup E|X_n| < \infty$, so that $X \in L^1$ (in particular, finite a.e.). □

Remark. The convergence in the theorem is a.e., *not* in L^1. Counterexample: let (Z_k) be i.i.d. with $P(Z_k = 0) = P(Z_k = 2) = 1/2$ (so that $E(Z_k) = 1$) and $M_n = \prod_{k=1}^n Z_k$. Then (M_n) is a martingale with respect to $\widetilde{\mathcal{F}}_n = \sigma(Z_1, \dots, Z_n)$:

$$E(M_{n+1} \mid \widetilde{\mathcal{F}}_n) = M_n\, E(Z_{n+1} \mid \widetilde{\mathcal{F}}_n) = M_n\, E(Z_{n+1}) = M_n,$$

and $E(M_n) = 1$ for every n, with $\sup_n E|M_n| = 1 < \infty$. By the theorem, M_n converges a.e., and the limit is 0: as soon as a $Z_k = 0$ comes up (which happens with probability 1), the product vanishes forever. But then

$$E|M_n - 0| = E(M_n) = 1 \not\to 0 :$$

there is no convergence in L^1, and the expectation is lost in the limit ($1 = \lim_n E(M_n) \ne E(\lim_n M_n) = 0$).

Theorem 16.5 – L^p convergence, $p > 1$

If $\sup_n E|X_n|^p < \infty$ with $p > 1$, then $X_n \to X$ also in L^p, and $X_n = E(X \mid \widetilde{\mathcal{F}}_n)$.

Proof. Since $L^p \subseteq L^1$ in finite measure (P is a probability), $\sup E|X_n| < \infty$ and by the previous theorem $X_n \to X$ a.e. with $X \in L^1$.

Uniform bound in L^p via Doob's inequality. Applying Doob's maximal inequality to the positive martingale $|X_n|^p$ (which is a submartingale by conditional Jensen with $\varphi(t) = |t|^p$), one obtains the L^p *maximal inequality*:

$$E\left(\sup_{k\le n} |X_k|^p\right) \le \left(\frac{p}{p-1}\right)^p \sup_{k\le n} E|X_k|^p.$$

Letting $n \to \infty$ and by monotone convergence, $E(\sup_n |X_n|^p) \le (p/(p-1))^p \sup_n E|X_n|^p < \infty$. Call $M = \sup_n |X_n|$, which satisfies $M \in L^p$.

Convergence in L^p. One has $|X_n - X|^p \le 2^p(|X_n|^p + |X|^p) \le 2^p(M^p + M^p) = 2^{p+1}M^p$ (because $|X| \le M$ in the limit). Since $X_n \to X$ a.e. and $|X_n - X|^p \le 2^{p+1}M^p \in L^1$, by dominated convergence $E|X_n - X|^p \to 0$.

Identity $X_n = E(X \mid \widetilde{\mathcal{F}}_n)$. For $A \in \widetilde{\mathcal{F}}_n$ and $m \ge n$, by the martingale property $\int_A X_n dP = \int_A X_m dP$. By convergence in L^1, $\int_A X_m dP \to \int_A X dP$. Hence $\int_A X_n dP = \int_A X dP$, and since X_n is $\widetilde{\mathcal{F}}_n$-measurable, by uniqueness $X_n = E(X \mid \widetilde{\mathcal{F}}_n)$. □

17 Brownian Motion

The passage from discrete to continuous time leads to *Brownian motion*, the limit of rescaled random walks. Before the axiomatic definition, the random walk is studied in some detail: its explicit computations are what motivate the properties of the limit process.

17.1 The simple random walk

Let $(X_n)_{n\geq 1}$ be i.i.d. with $P(X_n = 1) = P(X_n = -1) = 1/2$, and $S_n = X_1 + \cdots + X_n$ (with $S_0 = 0$): a particle starts at the origin and at each step moves one unit to the right or to the left according to a fair coin. The basic computations: $E(X_i) = 0$ and $\mathrm{Var}(X_i) = E(X_i^2) = 1$, whence

$$E(S_n) = 0, \qquad \mathrm{Var}(S_n) = E(S_n^2) = \sum_{i,j} E(X_i X_j) = \sum_{i=1}^{n} E(X_i^2) = n,$$

using that for $i \neq j$, $E(X_iX_j) = E(X_i)E(X_j) = 0$ by independence. Moreover, for $m < n$, $S_n - S_m = \sum_{i=m+1}^{n} X_i$ is another random walk, independent of $S_1, \ldots, S_m$, with $\mathrm{Var}(S_n - S_m) = n - m$.

Example 17.1 – Distribution of S_n and returns to the origin

After n steps, $S_n = a$ requires k steps $+1$ and $n-k$ steps -1 with $a = k-(n-k) = 2k-n$; in particular a and n have the same parity, and $k = (n+a)/2$. Of the 2^n equiprobable paths, the favorable ones are the choices of where to place the $+1$ steps:

$$P(S_n = a) = \frac{1}{2^n}\binom{n}{\frac{n+a}{2}}.$$

In particular, the probability of being at the origin after $2m$ steps is $P(S_{2m} = 0) = \binom{2m}{m}/2^{2m}$, which with Stirling's formula $n! \approx n^n e^{-n}\sqrt{2\pi n}$ is estimated as

$$P(S_{2m} = 0) = \frac{(2m)!}{(m!)^2\, 2^{2m}} \approx \frac{(2m)^{2m}e^{-2m}\sqrt{4\pi m}}{m^{2m}e^{-m}\cdot m^m e^{-m}\cdot(2\pi m)\cdot 2^{2m}} \cdot \frac{m^m}{m^m} = \frac{\sqrt{4\pi m}}{2\pi m} = \frac{1}{\sqrt{\pi m}}.$$

It decays, but very slowly: if $V = \sum_{m\geq 1} \mathbf{1}_{\{S_{2m}=0\}}$ is the number of returns to the origin,

$$E(V) = \sum_{m=1}^{\infty} P(S_{2m} = 0) \approx \sum_{m=1}^{\infty} \frac{1}{\sqrt{\pi m}} = \infty.$$

(Borel-Cantelli (II) is not applicable here: the events $\{S_{2m} = 0\}$ are not independent.)

Remark. $E(V) = \infty$ does not by itself imply $V = \infty$ a.e.: a variable can have infinite expectation and be finite with probability 1. This is the *St. Petersburg paradox*: a coin is tossed until the first tails, and if k heads came up before, one wins 2^k pesos; the expected gain is

$$E(G) = \sum_{k=0}^{\infty} 2^k \cdot \frac{1}{2^{k+1}} = \sum_{k=0}^{\infty} \frac{1}{2} = \infty,$$

but G is finite with probability 1. For the random walk the correct argument is a different one. Let ρ be the probability of ever returning to the origin. Each time the walk returns, the

process starts afresh (the later steps are independent and have the same distribution), so that $P(V \geq k) = \rho^k$. If $\rho < 1$,

$$E(V) = \sum_{k\geq 1} P(V \geq k) = \sum_{k\geq 1} \rho^k = \frac{\rho}{1-\rho} < \infty,$$

contradicting $E(V) = \infty$. Therefore $\rho = 1$: the random walk returns to the origin (and, iterating, visits it *infinitely many times*) with probability 1.

Example 17.2 – The Galton board and the Gaussian bell curve

On a Galton board, small balls fall through n levels of pegs; each ball performs a random walk of n steps, and the column over position a accumulates a fraction of balls proportional to $\binom{n}{(n+a)/2}$. Measuring each column in units of the central column, the height over a is

$$H_a = \binom{n}{\frac{n+a}{2}} \Big/ \binom{n}{\frac{n}{2}} = \frac{(n/2)!\,(n/2)!}{\left(\frac{n+a}{2}\right)!\left(\frac{n-a}{2}\right)!}.$$

Write $\rho = n/2$ and $c = a/2$; cancelling factorials and taking logarithms,

$$H_a = \frac{(\rho - c + 1)\cdots\rho}{(\rho+1)\cdots(\rho+c)}, \quad \ln H_a = \sum_{k=\rho-c+1}^{\rho} \ln k - \sum_{k=\rho+1}^{\rho+c} \ln k \approx \int_{\rho-c}^{\rho} \ln x\,dx - \int_{\rho}^{\rho+c} \ln x\,dx.$$

With the antiderivative $x\ln x - x$,

$$\ln H_a \approx 2\rho\ln\rho - (\rho-c)\ln(\rho-c) - (\rho+c)\ln(\rho+c).$$

The Taylor polynomial of order 2 of $f(x) = x\ln x$ at ρ (with $f'(x) = \ln x + 1$ and $f''(x) = 1/x$) gives $f(\rho \pm c) \approx \rho\ln\rho \pm (\ln\rho + 1)c + \frac{c^2}{2\rho}$; on substituting, the terms of order 0 and 1 cancel and one gets

$$\ln H_a \approx -\frac{c^2}{\rho} = -\frac{a^2/4}{n/2} = -\frac{a^2}{2n} = -\frac{1}{2}\left(\frac{a}{\sqrt{n}}\right)^2, \quad \text{that is} \quad H_a \approx e^{-x^2/2} \quad \text{with } x = \frac{a}{\sqrt{n}}.$$

Rescaling the horizontal axis by $\sqrt{n}$, the profiles fit the Gaussian bell curve: this is the *de Moivre-Laplace* theorem,

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1), \qquad P\left(\alpha \leq \frac{S_n}{\sqrt{n}} \leq \beta\right) \approx \frac{1}{\sqrt{2\pi}} \int_{\alpha}^{\beta} e^{-x^2/2}\,dx.$$

17.2 From the random walk to Brownian motion

Heuristically, Brownian motion is a random walk with time steps of length $\Delta t = 1/N$ and space steps of length ϵ, in the limit $N \to \infty$: one defines $W_{n/N} = \epsilon S_n$ and normalizes by requiring $\mathrm{Var}(W_1) = 1$. Since N steps are taken to reach $t = 1$,

$$1 = E(W_1^2) = \epsilon^2 E(S_N^2) = \epsilon^2 N \implies \epsilon = \frac{1}{\sqrt{N}}.$$

With that choice, for $t = n/N$,

$$W_t = \frac{S_n}{\sqrt{N}} = \frac{S_n}{\sqrt{n}} \cdot \sqrt{\frac{n}{N}} = \frac{S_n}{\sqrt{n}} \cdot \sqrt{t},$$

and since $S_n/\sqrt{n} \sim \mathcal{N}(0,1)$ for large n (de Moivre-Laplace), one expects $W_t \sim \mathcal{N}(0,t)$. From $\mathrm{Var}(S_n - S_m) = n - m$ and from the independence of the stretches of the random walk one also expects $W_t - W_s \sim \mathcal{N}(0, t-s)$ and independent increments. These properties, together with the continuity of the paths, are exactly the definition.

17.3 Definition and properties

Definition 17.1 – Brownian Motion

A *Brownian motion* (Wiener process) is a stochastic process $(W_t)_{t\geq 0}$ that satisfies:

(i) $W_0 = 0$ almost everywhere.

(ii) For $0 \leq s < t$, $W_t - W_s \sim \mathcal{N}(0, t-s)$.

(iii) Independent increments: for $0 \leq t_0 < t_1 < \cdots < t_n$, the variables $W_{t_1} - W_{t_0}, W_{t_2} - W_{t_1}, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent.

(iv) $t \mapsto W_t$ is continuous almost everywhere.

Remark. Existence (Wiener, 1923): such a process exists. The construction uses, for example, Kolmogorov's extension or a random series expansion.

Remark. The paths $t \mapsto W_t(\omega)$ are continuous but, almost surely, they are not differentiable at any point. The heuristics for both facts lies in the scale of the increments: since $\mathrm{Var}(W_{t+h} - W_t) = h$, the typical increment is $|W_{t+h} - W_t| \approx \sqrt{h}$, which tends to 0 (continuity) but whose difference quotient $\sqrt{h}/h = 1/\sqrt{h} \to \infty$ (non-differentiability). Moreover: a derivative at t_0 could be estimated with past increments ($r < t_0$) and would determine the nearby future increments, but $W_s - W_{t_0}$ is *independent* of $\{W_r : r \leq t_0\}$: the process cannot remember a slope. This is what makes classical calculus fail and forces the construction of the Itô integral.

Remark. Transformations that preserve Brownian motion: if (W_t) is a Brownian motion, so are

$$(c\,W_{t/c^2})_{t\geq 0} \quad \text{(self-similarity)}, \qquad (W_{t+s} - W_s)_{t\geq 0} \quad \text{(restart)},$$
$$(t\,W_{1/t})_{t\geq 0} \quad \text{(time inversion)}.$$

With drift and scale, $(\mu t + \sigma W_t)$ is called *Brownian motion with drift* and $(Ce^{\mu t + \sigma W_t})$ *geometric Brownian motion*.

17.4 Quadratic variation

Theorem 17.1 – Quadratic variation of Brownian motion

Let $a = t_0 < t_1 < \cdots < t_m = b$ be a partition of $[a,b]$ with mesh $\delta = \max_k(t_{k+1} - t_k)$. Define

$$Q_m = \sum_{k=1}^{m} (W_{t_k} - W_{t_{k-1}})^2.$$

Then $Q_m \xrightarrow{L^2} b - a$ as $\delta \to 0$.

Proof. Call $\Delta_k = W_{t_k} - W_{t_{k-1}}$ and $\Delta t_k = t_k - t_{k-1}$. By definition of Brownian motion, $\Delta_k \sim \mathcal{N}(0, \Delta t_k)$ and the Δ_k are independent.

Expectation. $E(\Delta_k^2) = \mathrm{Var}(\Delta_k) = \Delta t_k$, so that

$$E(Q_m) = \sum_{k=1}^{m} E(\Delta_k^2) = \sum_{k=1}^{m} \Delta t_k = b - a.$$

Variance. By independence of the Δ_k^2,

$$\mathrm{Var}(Q_m) = \sum_{k=1}^{m} \mathrm{Var}(\Delta_k^2).$$

Compute $\mathrm{Var}(\Delta_k^2)$: if $Z \sim \mathcal{N}(0, \sigma^2)$, then $Z^2 = \sigma^2 \xi$ with $\xi \sim \chi_1^2$. Since $E(\xi^2) = 3$ (fourth moment of the standard normal) and $E(\xi) = 1$, $\mathrm{Var}(\xi) = 2$. Hence $\mathrm{Var}(Z^2) = \sigma^4 \cdot 2 = 2\sigma^4$, and in this case $\mathrm{Var}(\Delta_k^2) = 2(\Delta t_k)^2$.

Substituting,

$$\mathrm{Var}(Q_m) = 2\sum_{k=1}^{m} (\Delta t_k)^2 \leq 2\delta \sum_{k=1}^{m} \Delta t_k = 2\delta(b-a) \xrightarrow{\delta \to 0} 0.$$

Conclusion. $E((Q_m - (b-a))^2) = \mathrm{Var}(Q_m) \to 0$, that is $Q_m \to b - a$ in L^2. □

Remark. Heuristically, this is written as $(dW)^2 = dt$. In the limit one also has

$$\sum (\Delta t)^2 \to 0, \qquad \sum \Delta t \, \Delta W \to 0, \qquad \sum (\Delta W)^2 \to b - a,$$

which are the rules used when constructing the Itô integral and the corresponding lemma.

18 Itô Integral

One would like to define $\int_a^b X_t\,dW_t$ following Riemann: approximate by sums $\sum X_{\xi_k}(W_{t_{k+1}}-W_{t_k})$ with $\xi_k \in [t_k, t_{k+1}]$. But unlike the classical case, the choice of ξ_k *does* matter, because the paths of Brownian motion do not have bounded variation. The choice that gives an integral with good martingale properties is the *left endpoint*: $\xi_k = t_k$.

Definition 18.1 – Itô Integral

Let $(X_t)_{t\in[a,b]}$ be an adapted process with $X_\cdot \in L^2([a,b]\times\Omega)$ (sufficiently integrable). For a partition $a = t_0 < t_1 < \cdots < t_m = b$ define the *left-endpoint Riemann sum*

$$R_m = \sum_{k=1}^{m} X_{t_{k-1}}(W_{t_k} - W_{t_{k-1}}).$$

The Itô integral is defined as

$$\int_a^b X\,dW = \lim_{\delta\to 0} R_m,$$

in $L^2(\Omega)$. The limit exists and does not depend on the sequence of partitions.

Remark. The choice of the left endpoint is essential. Taking $\xi_k = t_k$ (right endpoint) or $\xi_k = (t_{k-1}+t_k)/2$ (midpoint, Stratonovich) yields *different* integrals. Itô's is the only one that keeps the martingale property. The difference is computed directly for $\int_a^b W\,dW$: if B_m uses the right endpoint and A_m the left one,

$$B_m - A_m = \sum_k \left(W_{t_k} - W_{t_{k-1}}\right)\left(W_{t_k} - W_{t_{k-1}}\right) = Q_m \xrightarrow{L^2} b - a \neq 0$$

by the quadratic variation: both limits exist, but they differ by exactly $b-a$.

18.1 Properties of the Itô integral

Proposition 18.1 – Properties

Let X, Y be adapted processes that are Itô integrable, and $\alpha, \beta \in \mathbb{R}$:

(i) $\int_a^b(\alpha X + \beta Y)dW = \alpha\int_a^b X\,dW + \beta\int_a^b Y\,dW$.

(ii) $E\left[\int_a^b X\,dW\right] = 0$.

(iii) $E\left[\left(\int_a^b X\,dW\right)^2\right] = E\left[\int_a^b X^2\,dt\right]$.

(iv) $E\left[\int_a^b X\,dW \cdot \int_a^b Y\,dW\right] = E\left[\int_a^b XY\,dt\right]$.

Proof. (ii) and (iii) are proved at the level of the left-endpoint Riemann sums (the general case follows by passing to the limit in L^2). Write $R = \sum_k X_{t_{k-1}}\Delta W_k$ with $\Delta W_k = W_{t_k} - W_{t_{k-1}}$ and $\Delta t_k = t_k - t_{k-1}$.

(ii) Since $X_{t_{k-1}}$ is $\mathcal{F}_{t_{k-1}}$-measurable (the process is adapted) and ΔW_k is independent of $\mathcal{F}_{t_{k-1}}$ with mean 0,

$$E(X_{t_{k-1}}\Delta W_k) = E(X_{t_{k-1}})\,E(\Delta W_k) = 0,$$

and summing, $E(R) = 0$.

(iii) Expanding the square,

$$E(R^2) = \sum_k E\big(X_{t_{k-1}}^2 (\Delta W_k)^2\big) + 2\sum_{j<k} E\big(X_{t_{j-1}} \Delta W_j \, X_{t_{k-1}} \Delta W_k\big).$$

In the cross terms with $j < k$, the factor $X_{t_{j-1}} \Delta W_j X_{t_{k-1}}$ is $\mathcal{F}_{t_{k-1}}$-measurable, and ΔW_k is independent of it and centered: each cross term vanishes. In the diagonal ones, $(\Delta W_k)^2$ is independent of $X_{t_{k-1}}^2$ and $E((\Delta W_k)^2) = \Delta t_k$, whence

$$E(R^2) = \sum_k E(X_{t_{k-1}}^2)\, \Delta t_k = E\left(\sum_k X_{t_{k-1}}^2 \, \Delta t_k\right) \xrightarrow{\delta \to 0} E\left(\int_a^b X^2 \, dt\right).$$

Property (i) is the linearity of the Riemann sums, and (iv) follows from (iii) by polarization, $XY = \frac{1}{4}\big[(X+Y)^2 - (X-Y)^2\big]$. □

Remark. The Itô isometry is the formal change $(dW)^2 \mapsto dt$ mentioned earlier: the square of the stochastic integral translates into the classical integral of X^2 with respect to time.

18.2 Itô's Lemma

The change of variables in stochastic calculus is not the classical one: an additional term appears due to $(dW)^2 = dt$.

Remark. (*Orders of magnitude.*) The reason why the second-order term survives is seen by comparing, over a partition of $[a, b]$ with mesh δ, the size of the four differentials that appear when expanding by Taylor:

$$\begin{aligned}
dt: &\qquad \sum_k (t_k - t_{k-1}) = b - a, \\
(dW)^2: &\qquad \sum_k (W_{t_k} - W_{t_{k-1}})^2 \xrightarrow{L^2} b - a \qquad \text{(quadratic variation)}, \\
(dt)^2: &\qquad \sum_k (t_k - t_{k-1})^2 \le \delta\,(b-a) \longrightarrow 0, \\
dt\,dW: &\qquad \sum_k |W_{t_k} - W_{t_{k-1}}|\,(t_k - t_{k-1}) \approx \sqrt{\delta}\,(b-a) \longrightarrow 0,
\end{aligned}$$

where the last line used that $E((W_{t_k} - W_{t_{k-1}})^2) = t_k - t_{k-1}$, so that the typical increment is $|W_{t_k} - W_{t_{k-1}}| \approx \sqrt{t_k - t_{k-1}} \le \sqrt{\delta}$. That is: dt and $(dW)^2$ are of the same order and contribute, whereas $(dt)^2$ and $dt\,dW$ vanish in the limit. In classical calculus (where $(dt)^2$ appears instead of $(dW)^2$) the second-order term disappears and the change of variables keeps a single integral.

Theorem 18.1 – Itô's Lemma

Let $g \in C^2(\mathbb{R})$. If $Z_t = g(W_t)$,

$$dZ_t = g'(W_t)\, dW_t + \tfrac{1}{2} g''(W_t)\, dt.$$

In integral form:

$$g(W_b) - g(W_a) = \int_a^b g'(W_s)\, dW_s + \frac{1}{2}\int_a^b g''(W_s)\, ds.$$

Proof. Fix a partition $a = t_0 < t_1 < \cdots < t_m = b$ with mesh $\delta = \max_k(t_k - t_{k-1})$. Write $\Delta W_k = W_{t_k} - W_{t_{k-1}}$ and $\Delta t_k = t_k - t_{k-1}$. By a telescoping sum,

$$g(W_b) - g(W_a) = \sum_{k=1}^{m} [g(W_{t_k}) - g(W_{t_{k-1}})].$$

Taylor of order 2. Since $g \in C^2$, for each k there is ξ_k between $W_{t_{k-1}}$ and W_{t_k} with

$$g(W_{t_k}) - g(W_{t_{k-1}}) = g'(W_{t_{k-1}})\Delta W_k + \tfrac{1}{2} g''(\xi_k)(\Delta W_k)^2.$$

Summing over k, the sum splits into two terms:

$$g(W_b) - g(W_a) = \underbrace{\sum_{k=1}^{m} g'(W_{t_{k-1}})\Delta W_k}_{I_m} + \underbrace{\tfrac{1}{2}\sum_{k=1}^{m} g''(\xi_k)(\Delta W_k)^2}_{J_m}.$$

Term I_m. By the construction of the Itô integral as the L^2 limit of left-endpoint Riemann sums,

$$I_m \xrightarrow{L^2(\Omega)} \int_a^b g'(W_s)\, dW_s.$$

Term J_m. It is split in two:

$$J_m = \tfrac{1}{2}\sum_k g''(W_{t_{k-1}})(\Delta W_k)^2 + \tfrac{1}{2}\sum_k [g''(\xi_k) - g''(W_{t_{k-1}})](\Delta W_k)^2 =: J_m^{(1)} + J_m^{(2)}.$$

For $J_m^{(1)}$, the quadratic variation is used: given $\epsilon > 0$, fine partitions make $\sum_k (\Delta W_k)^2 \to b - a$ in L^2. Moreover, for a continuous function h, the convergence

$$\sum_k h(W_{t_{k-1}})(\Delta W_k)^2 \xrightarrow{L^2} \int_a^b h(W_s)\, ds$$

holds. (This result is the analogue of the fundamental theorem for quadratic variation.) Applying it with $h = g''$,

$$J_m^{(1)} \xrightarrow{L^2} \tfrac{1}{2}\int_a^b g''(W_s)\, ds.$$

For $J_m^{(2)}$: since g'' is continuous and the ξ_k lie between $W_{t_{k-1}}$ and W_{t_k} with $|W_{t_k} - W_{t_{k-1}}| \to 0$ by uniform continuity of Brownian motion on compact sets, $\max_k |g''(\xi_k) - g''(W_{t_{k-1}})| \to 0$ almost everywhere. Since $\sum (\Delta W_k)^2$ is bounded in L^1 (its expectation is $b - a$),

$$|J_m^{(2)}| \le \tfrac{1}{2}\max_k |g''(\xi_k) - g''(W_{t_{k-1}})| \cdot \sum_k (\Delta W_k)^2 \to 0.$$

Conclusion. Letting $\delta \to 0$ in $g(W_b) - g(W_a) = I_m + J_m^{(1)} + J_m^{(2)}$, one gets

$$g(W_b) - g(W_a) = \int_a^b g'(W_s)\, dW_s + \tfrac{1}{2}\int_a^b g''(W_s)\, ds.$$

□

Version with t. If $g \in C^{1,2}(\mathbb{R}_{\ge 0} \times \mathbb{R})$ and $Z_t = g(t, W_t)$,

$$dZ_t = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial x} dW_t + \frac{1}{2}\frac{\partial^2 g}{\partial x^2} dt.$$

18.3 Examples

Example 18.1 – Square of Brownian motion

Applying Itô to $g(x) = x^2$, $g'(x) = 2x$, $g''(x) = 2$. One gets

$$W_b^2 - W_a^2 = \int_a^b 2W_s\, dW_s + \int_a^b 1\, ds,$$

whence $\int_a^b W\, dW = \frac{W_b^2 - W_a^2}{2} - \frac{b-a}{2}$. Note the term $-(b-a)/2$, absent in classical calculus.

Example 18.2 – Geometric Brownian motion

Solve the stochastic differential equation

$$dX_t = \mu X_t\, dt + \sigma X_t\, dW_t, \qquad X_0 \text{ given.}$$

Take $Y_t = \ln X_t$ and apply Itô with $g(x) = \ln x$, $g'(x) = 1/x$, $g''(x) = -1/x^2$:

$$\begin{aligned} d(\ln X_t) &= \frac{1}{X_t} dX_t - \frac{1}{2X_t^2}(dX_t)^2 \\ &= \frac{1}{X_t}(\mu X_t\, dt + \sigma X_t\, dW_t) - \frac{1}{2X_t^2}\sigma^2 X_t^2\, dt \\ &= \left(\mu - \tfrac{\sigma^2}{2}\right) dt + \sigma\, dW_t. \end{aligned}$$

Integrating, $\ln X_t = \ln X_0 + (\mu - \frac{\sigma^2}{2})t + \sigma W_t$, so that

$$X_t = X_0 \exp\!\left(\sigma W_t + \left(\mu - \tfrac{\sigma^2}{2}\right) t\right).$$

This is the classical model of the price of an asset in finance (Black-Scholes model).

18.4 General stochastic processes

Definition 18.2 – Itô process

An *Itô process* is a process of the form

$$X_t = X_0 + \int_0^t f(X_s, s)\, ds + \int_0^t g(X_s, s)\, dW_s,$$

written in differential form as

$$dX_t = f(X_t, t)\, dt + g(X_t, t)\, dW_t.$$

f is called the *drift* and g the *diffusion coefficient*.

Theorem 18.2 – Generalized Itô

If X_t is an Itô process as above and $h \in C^{1,2}(\mathbb{R}_{\geq 0} \times \mathbb{R})$, $M_t = h(t, X_t)$ satisfies

$$dM_t = \left(\frac{\partial h}{\partial t} + f \frac{\partial h}{\partial x} + \frac{g^2}{2} \frac{\partial^2 h}{\partial x^2} \right) dt + g \frac{\partial h}{\partial x}\, dW_t.$$

Proof. Same strategy as Itô's lemma. Partition $[0, T]$ with $0 = t_0 < t_1 < \cdots < t_m = T$ and apply Taylor of order 2 to h:

$$h(t_k, X_{t_k}) - h(t_{k-1}, X_{t_{k-1}}) = h_t \Delta t_k + h_x \Delta X_k + \tfrac{1}{2}[h_{tt}(\Delta t_k)^2 + 2h_{tx}\Delta t_k \Delta X_k + h_{xx}(\Delta X_k)^2] + \cdots,$$

with derivatives evaluated at $(t_{k-1}, X_{t_{k-1}})$. Since $\Delta X_k = f\Delta t_k + g\Delta W_k + o(\sqrt{\Delta t_k})$,

$$(\Delta X_k)^2 = g^2(\Delta W_k)^2 + 2fg\Delta t_k \Delta W_k + f^2(\Delta t_k)^2 + \cdots.$$

The asymptotic rules of quadratic variation give

$$\sum (\Delta t_k)^2 \to 0, \qquad \sum \Delta t_k \Delta W_k \to 0, \qquad \sum (\Delta W_k)^2 \to \int dt,$$

so that only the term $h_{xx} g^2 (\Delta W)^2/2 \to \frac{g^2}{2} h_{xx}\, dt$ survives. Adding the first-order terms, $h_t\, dt + h_x(f\, dt + g\, dW) = (h_t + f h_x)dt + g h_x\, dW$. Adding the surviving quadratic term of h_{xx}, the stated formula is obtained. □

19 Girsanov's Theorem

What happens to the distribution of a process if the probability measure is changed? Girsanov says exactly how a Brownian motion transforms under an absolutely continuous change of measure.

19.1 Motivation: change of mean of a normal

Let $Z \sim \mathcal{N}(0,1)$ under P. It is shown that there is a measure Q, equivalent to P (that is, $P \ll Q \ll P$), such that $Z \sim \mathcal{N}(\mu, 1)$ under Q. For this, first compute

$$E_P(e^{\mu Z}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\mu t} e^{-t^2/2}\, dt = \frac{1}{\sqrt{2\pi}} \int e^{-(t-\mu)^2/2} dt \cdot e^{\mu^2/2} = e^{\mu^2/2}.$$

Therefore $E_P(e^{\mu Z - \mu^2/2}) = 1$.

Change of measure. For $A \in \mathcal{F}$ define

$$Q(A) = \int_A e^{\mu Z - \mu^2/2}\, dP.$$

Q is a probability measure (positive, σ-additive, $Q(\Omega) = 1$). Under Q, the distribution of Z is computed:

$$\begin{aligned} F_Q(a) = Q(\{Z \le a\}) &= \int_{\{Z \le a\}} e^{\mu Z - \mu^2/2}\, dP \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathbf{1}_{\{t \le a\}} e^{\mu t - \mu^2/2} e^{-t^2/2} dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{a} e^{-(t-\mu)^2/2} dt, \end{aligned}$$

that is, $Z \sim \mathcal{N}(\mu, 1)$ under Q.

19.2 Continuous version

Before the statement, observe why the candidate density is a martingale. Let

$$Y_t = \int_0^t X_s\, dW_s - \frac{1}{2} \int_0^t X_s^2\, ds, \qquad M_t = e^{Y_t},$$

that is $dY_t = X_t\, dW_t - \frac{1}{2} X_t^2\, dt$, with $(dY_t)^2 = X_t^2\, dt$ by the rules of quadratic variation. By Itô's lemma with $g(s) = e^s$,

$$dM_t = e^{Y_t}\, dY_t + \tfrac{1}{2} e^{Y_t} (dY_t)^2 = e^{Y_t} \left(X_t\, dW_t - \tfrac{1}{2} X_t^2\, dt + \tfrac{1}{2} X_t^2\, dt \right) = M_t X_t\, dW_t.$$

The drift term cancels exactly (that is what the $-\frac{1}{2}\int_0^t X_s^2\, ds$ in the definition is for), and one gets $M_t = 1 + \int_0^t M_s X_s\, dW_s$: an Itô integral, which is a martingale with mean $E(M_t) = 1$ provided the integrand is sufficiently integrable. The Novikov condition in the statement guarantees exactly that.

Theorem 19.1 – Girsanov

Let $(W_t)_{t\in[0,T]}$ be a Brownian motion under P and let (X_t) be a sufficiently integrable adapted process. Define

$$M_t = \exp\left(\int_0^t X_s\,dW_s - \frac{1}{2}\int_0^t X_s^2\,ds\right).$$

Under reasonable conditions (*Novikov's condition*: $E[e^{\frac{1}{2}\int_0^T X_s^2 ds}] < \infty$), M_t is a martingale and $E(M_T) = 1$. Define the measure

$$Q(A) = \int_A M_T\,dP \qquad (A \in \mathcal{F}_T).$$

Then Q is equivalent to P, and the process

$$\widetilde{W}_t = W_t - \int_0^t X_s\,ds$$

is a Brownian motion under Q.

Remark. In the particular case $X_t \equiv \mu/\sigma$ (constant), taking $W'_t = \sigma W_t$ with drift μ, Girsanov says that $\widetilde{W}_t = W_t - \mu t/\sigma$ is a Brownian motion under Q with $dQ/dP = \exp(\frac{\mu}{\sigma}W_T - \frac{\mu^2}{2\sigma^2}T)$. This justifies the change of the geometric Black-Scholes model from the real-world measure to the risk-neutral measure, on which the pricing of derivatives rests.

Remark. Girsanov is the continuous version of Radon-Nikodym applied to processes. The derivative $dQ/dP = M_T$ is called the *Doléans-Dade exponential* of $\int X\,dW$.

20 Application: the Black-Scholes model

The tools constructed so far (Brownian motion, Itô, Girsanov) have an emblematic application: the Black-Scholes formula for pricing financial options. It is a case where the whole apparatus is put to work.

20.1 Model of the asset

Consider an *underlying asset* with price S_t that evolves according to a geometric Brownian motion:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t, \qquad S_0 \text{ given},$$

where $\mu \in \mathbb{R}$ is the *expected rate of return* and $\sigma > 0$ the *volatility*. As seen when solving the SDE with Itô,

$$S_t = S_0 \exp\Big(\sigma W_t + \Big(\mu - \tfrac{\sigma^2}{2}\Big)\,t\Big).$$

Alongside there is a *risk-free asset* (bond) that grows at a constant rate r:

$$B_t = e^{rt}, \qquad dB_t = rB_t\,dt.$$

Derivative. A *derivative* is a contract that pays a function $\Phi(S_T)$ at a future time T. For example, a European *call option* with *strike* K pays

$$\Phi(S_T) = (S_T - K)^+ = \max(S_T - K, 0).$$

The problem: what is the fair price C_0 to pay today for this contract?

20.2 No-arbitrage argument and risk-neutral measure

Strategy. If one can build, with C_0 dollars today, a portfolio that at T replicates exactly the payoff $\Phi(S_T)$, then C_0 is the only price compatible with the *absence of arbitrage* (no possibility of a sure gain without investment).

Reduction to the discounted asset. Let $\widetilde{S}_t = S_t/B_t = e^{-rt}S_t$, the price of the asset measured in bonds. By Itô,

$$d\widetilde{S}_t = (\mu - r)\widetilde{S}_t\,dt + \sigma\widetilde{S}_t\,dW_t.$$

If the drift $(\mu - r)\widetilde{S}_t\,dt$ could be made to disappear, $\widetilde{S}$ would be a martingale. This is exactly what Girsanov provides: taking

$$X_t \equiv \frac{\mu - r}{\sigma},$$

the measure Q with $dQ/dP = \exp(-\frac{\mu-r}{\sigma}W_T - \frac{1}{2}(\frac{\mu-r}{\sigma})^2 T)$ makes

$$\widetilde{W}_t = W_t + \tfrac{\mu-r}{\sigma}t$$

a Brownian motion under Q. Rewriting,

$$d\widetilde{S}_t = \sigma\widetilde{S}_t\,d\widetilde{W}_t \qquad (\text{under } Q),$$

that is, $\widetilde{S}$ is a martingale under Q.

Definition 20.1 – Risk-neutral measure

The *risk-neutral measure* is the unique measure Q equivalent to P such that $\widetilde{S}_t = e^{-rt}S_t$ is a martingale under Q. Under Q, the price of the asset evolves with drift r (not μ):

$$dS_t = rS_t\,dt + \sigma S_t\,d\widetilde{W}_t, \qquad S_t = S_0 \exp\Big(\sigma\widetilde{W}_t + (r - \tfrac{\sigma^2}{2})t\Big).$$

Remark. The name "risk-neutral" is because under Q *every asset* (both S and B) grows, on average, at the same rate r: there is no risk premium. The investors' attitude toward risk, which affects the real-world drift μ, disappears under Q.

20.3 Pricing formula

Theorem 20.1 – Black-Scholes price (general version)

Under no arbitrage and market completeness, the fair price at $t = 0$ of a derivative with payoff $\Phi(S_T)$ is

$$C_0 = e^{-rT}E_Q(\Phi(S_T)).$$

Proof. *Self-financing portfolio.* A portfolio is described by adapted processes (α_t, β_t) that indicate how many units of the asset and of the bond are held at each moment. Its value is

$$V_t = \alpha_t S_t + \beta_t B_t.$$

The portfolio is *self-financing* if its variation comes only from the price variations:

$$dV_t = \alpha_t\,dS_t + \beta_t\,dB_t.$$

In terms of the discounted value $\widetilde{V}_t = V_t/B_t$, by Itô applied to $V_t \cdot e^{-rt}$,

$$d\widetilde{V}_t = \alpha_t\,d\widetilde{S}_t.$$

Replication and martingale under Q. Suppose there is a self-financing portfolio that *replicates* the payoff: $V_T = \Phi(S_T)$. Under Q, $\widetilde{S}_t$ is a martingale (by Girsanov), and then

$$\widetilde{V}_t - \widetilde{V}_0 = \int_0^t \alpha_s\,d\widetilde{S}_s$$

is an Itô integral with respect to a martingale, therefore $\widetilde{V}_t$ is a martingale under Q (assuming the necessary integrability). Taking expectations,

$$\widetilde{V}_0 = E_Q(\widetilde{V}_T) = E_Q(e^{-rT}\Phi(S_T)).$$

And $V_0 = \widetilde{V}_0$ because $B_0 = 1$, which gives $C_0 = V_0 = e^{-rT}E_Q(\Phi(S_T))$.

Existence of the replicating portfolio. In the Black-Scholes model (one asset, one bond), the market is *complete*: by a martingale representation theorem (the discounted martingale $\widetilde{C}_t = E_Q(e^{-rT}\Phi(S_T) \mid \widetilde{\mathcal{F}}_t)$ is written as an Itô integral with respect to $\widetilde{W}$, which gives α_t), such a portfolio always exists. The *no-arbitrage* argument implies that its initial price is unique. □

20.4 Explicit formula for the European call

Take $\Phi(s) = (s-K)^+$. Under Q, $S_T = S_0 \exp(\sigma \widetilde{W}_T + (r - \sigma^2/2)T)$, with $\widetilde{W}_T \sim \mathcal{N}(0,T)$, that is $\ln S_T \sim \mathcal{N}(\ln S_0 + (r-\sigma^2/2)T,\ \sigma^2 T)$.

Define

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

and $N(\cdot)$ the distribution function of a standard normal.

Theorem 20.2 – Black-Scholes formula

The price of a European call option with strike K and maturity T is

$$C_0 = S_0\, N(d_1) - K e^{-rT}\, N(d_2).$$

Proof (direct computation). Compute $E_Q((S_T-K)^+)$. Under Q write

$$S_T = S_0\, e^{\sigma\sqrt{T}Z + (r-\sigma^2/2)T}, \qquad Z \sim \mathcal{N}(0,1).$$

The condition $S_T > K$ is equivalent to

$$Z > \frac{\ln(K/S_0) - (r-\sigma^2/2)T}{\sigma\sqrt{T}} = -d_2.$$

Therefore,

$$\begin{aligned} E_Q(S_T \mathbf{1}_{S_T>K}) &= S_0 e^{(r-\sigma^2/2)T} \int_{-d_2}^{\infty} e^{\sigma\sqrt{T}z} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\ &= S_0 e^{rT} \int_{-d_2}^{\infty} \frac{e^{-(z-\sigma\sqrt{T})^2/2}}{\sqrt{2\pi}} dz \qquad \text{(completing the square)} \\ &= S_0 e^{rT} \cdot P(Z' > -d_2 - \sigma\sqrt{T}) \quad \text{with } Z' \sim \mathcal{N}(0,1) \\ &= S_0 e^{rT}\, N(d_2 + \sigma\sqrt{T}) = S_0 e^{rT}\, N(d_1). \end{aligned}$$

And $E_Q(K\mathbf{1}_{S_T>K}) = K \cdot P(Z > -d_2) = K\, N(d_2)$. Substituting in $C_0 = e^{-rT} E_Q((S_T-K)^+) = e^{-rT}[E_Q(S_T\mathbf{1}_{S_T>K}) - K E_Q(\mathbf{1}_{S_T>K})]$,

$$C_0 = e^{-rT}[S_0 e^{rT} N(d_1) - K N(d_2)] = S_0 N(d_1) - K e^{-rT} N(d_2).$$

□

20.5 Financial interpretation

- $N(d_2)$ is the probability under Q that the option *ends in-the-money*, that is, $S_T > K$. The second term $Ke^{-rT}N(d_2)$ is the discounted value of paying K in that case.
- $S_0 N(d_1)$ is the *delta hedge*: the amount of underlying (in present value) that must be held in the portfolio to replicate the payoff of the call.
- The formula *does not depend* on μ, the real drift of the asset. It depends only on σ, r, S_0, K, T. This was a historical surprise: the fair price does not require estimating how the stock grows, only how much it fluctuates.

Remark. The model has strong assumptions (constant volatility, no jumps, no transaction costs, possibility of continuous trading). In practice systematic deviations are observed (volatility smile, jumps) that motivate more sophisticated models (Heston, models with jumps, stochastic volatility). But Black-Scholes is still the baseline reference.